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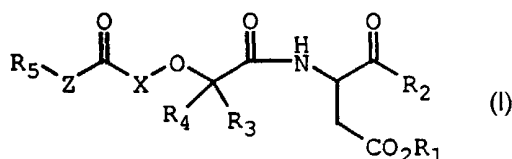
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WO 01/16093 A1

(54) Title: SUBSTITUTED α -HYDROXY ACID CASPASE INHIBITORS AND THE USE THEREOF



(57) Abstract: The present invention is directed to novel substituted α -hydroxy acid thereof, represented by general Formula (I), where R_1 - R_5 , X and Z are defined herein. The present invention also relates to the discovery that compounds having Formula (I) are potent inhibitors of caspases and apoptotic cell death. Therefore, the inhibitors of this invention can retard or block cell death in a variety of clinical conditions in which the loss of cells, tissues or entire organs occurs.

Substituted α -Hydroxy Acid Caspase Inhibitors and the Use Thereof

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Background of the Invention

Field of the Invention

10 This invention is in the field of medicinal chemistry. In particular, the invention relates to substituted α -hydroxy acids that are inhibitors of caspases. The invention also relates to the use of these substituted α -hydroxy acids for reducing or treating apoptotic cell death and/or reducing interleukin 1- β production.

15

Prior Art

Organisms eliminate unwanted cells by a process variously known as regulated cell death, programmed cell death or apoptosis. Such cell death occurs as a normal aspect of animal development as well as in tissue homeostasis and aging (Glucksmann, A., *Biol. Rev. Cambridge Philos. Soc.* 26:59-86 (1951); Glucksmann, A., *Archives de Biologie* 76:419-437 (1965); Ellis *et al.*, *Dev.* 112:591-603 (1991); Vaux *et al.*, *Cell* 76:777-779 (1994)). Apoptosis regulates cell number, facilitates morphogenesis, removes harmful or otherwise abnormal cells and eliminates cells that have already performed their function. Additionally, apoptosis occurs in response to various physiological stresses, such as hypoxia or ischemia (PCT published application WO96/20721).

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There are a number of morphological changes shared by cells experiencing regulated cell death, including plasma and nuclear membrane blebbing, cell shrinkage (condensation of nucleoplasm and cytoplasm), organelle relocation and compaction, chromatin condensation and

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production of apoptotic bodies (membrane enclosed particles containing intracellular material) (Orrenius, S., *J. Internal Medicine* 237:529-536 (1995)).

Apoptosis is achieved through an endogenous mechanism of cellular suicide (Wyllie, A. H., in *Cell Death in Biology and Pathology*, Bowen and Lockshin, eds., Chapman and Hall (1981), pp. 9-34). A cell activates its internally encoded suicide program as a result of either internal or external signals. The suicide program is executed through the activation of a carefully regulated genetic program (Wyllie *et al.*, *Int. Rev. Cyt.* 68:251 (1980); Ellis *et al.*, *Ann. Rev. Cell Bio.* 7:663 (1991)). Apoptotic cells and bodies are usually recognized and cleared by neighboring cells or macrophages before lysis. Because of this clearance mechanism, inflammation is not induced despite the clearance of great numbers of cells (Orrenius, S., *J. Internal Medicine* 237:529-536 (1995)).

Mammalian interleukin-1 β (IL-1 β) plays an important role in various pathologic processes, including chronic and acute inflammation and autoimmune diseases (Oppenheim, J. H. *et al.* *Immunology Today*, 7, 45-56 (1986)). IL-1 β is synthesized as a cell associated precursor polypeptide (pro-IL-1 β) that is unable to bind IL-1 receptors and is biologically inactive (Mosley *et al.*, *J. Biol. Chem.* 262:2941-2944 (1987)). By inhibiting conversion of precursor IL-1 β to mature IL-1 β , the activity of interleukin-1 can be inhibited. Interleukin-1 β converting enzyme (ICE) is a protease responsible for the activation of interleukin-1 β (IL-1 β) (Thornberry, N.A., *et al.*, *Nature* 356:768 (1992); Yuan, J., *et al.*, *Cell* 75:641 (1993)). ICE is a substrate-specific cysteine protease that cleaves the inactive prointerleukin-1 to produce the mature IL-1. The genes that encode for ICE and CPP32 are members of the mammalian ICE/Ced-3 family of genes which presently includes at least twelve members: ICE, CPP32/Yama/Apopain, mICE2, ICE4, ICH1, TX/ICH-2, MCH2, MCH3, MCH4, FLICE/MACH/MCH5, ICE-LAP6 and ICE_{rel}III. The proteolytic activity of this family of cysteine proteases, whose active site (a cysteine residue) is essential for ICE-mediated apoptosis, appears critical in mediating cell death (Miura *et al.*, *Cell* 75:653-660 (1993)).

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This gene family has recently been named caspases (Alnemri, E. S. et. al. *Cell*, 87, 171 (1996), and Thornberry, N. A. et. al. *J. Biol. Chem.* 272, 17907-17911 (1997)) and divided into three groups according to its known functions. Table 1 summarizes these known caspases.

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Table 1

Enzyme*
Group I: mediators of inflammation
Caspase-1 (ICE)
Caspase-4 (ICE _{rel} -II, TX, ICH-2)
Caspase-5 (ICE _{rel} -III, TY)
Group II: effectors of apoptosis
Caspase-2 (ICH-1, mNEDD2)
Caspase-3 (apopain, CPP-32, YAMA)
Caspase-7 (Mch-3, ICE-LAP3, CMH-1)
Group III: activators of apoptosis
Caspase-6 (Mch2)
Caspase-8 (MACH, FLICE, Mch5)
Caspase-9 (ICE-LAP6, Mch6)
Caspase-10

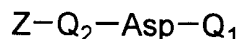
IL-1 is also a cytokine involved in mediating a wide range of biological responses including inflammation, septic shock, wound healing, hematopoiesis and growth of certain leukemias (Dinarello, C.A., *Blood* 77:1627-1652 (1991); diGiovine *et al.*, *Immunology Today* 11:13 (1990)).

Many potent caspase inhibitors have been prepared based on the peptide substrate structures of caspases. However, in contrast to their potency *in vitro*, there are few inhibitors with good efficacy ($IC_{50} < 1 \mu M$) in whole-cell models of apoptosis have been reported (Thornberry, N. A. *Chem. Biol.* 5:R97-103 (1998)). Therefore the need exists for cell death inhibitors that are efficacious in whole-cell models of apoptosis and active in animal model of apoptosis. These inhibitors thus can be employed as therapeutic agents to treat disease states in which regulated cell death and the cytokine activity of IL-1 play a role.

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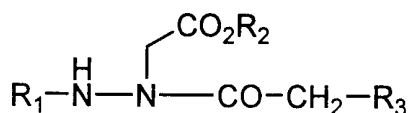
WO 93/05071 discloses peptide ICE inhibitors with the formula:

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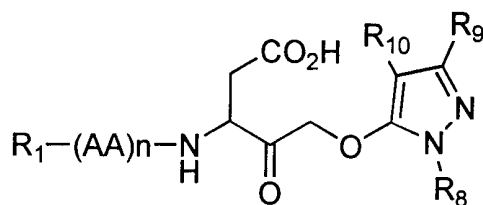
wherein Z is an N-terminal protecting group; Q₂ is 0 to 4 amino acids such that the sequence Q₂-Asp corresponds to at least a portion of the sequence Ala-Tyr-Val-His-Asp; Q₁ comprises an electronegative leaving group.

WO 96/03982 discloses aspartic acid analogs as ICE inhibitors with the formula:



wherein R₂ is H or alkyl; R₃ is a leaving group such as halogen; R₁ is heteroaryl-CO or an amino acid residue.

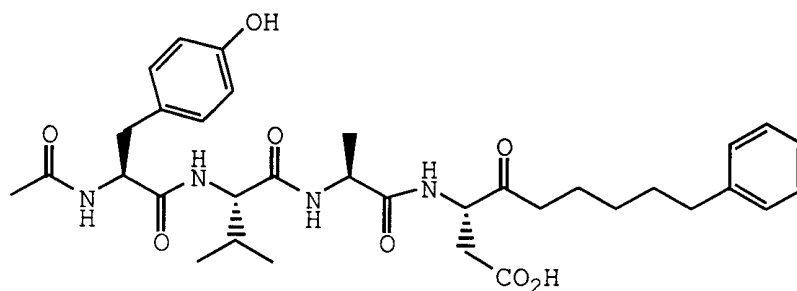
U.S. patent 5,585,357 discloses peptidic ketones as ICE inhibitors with the formula:



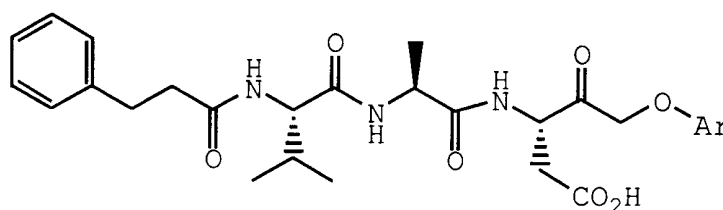
wherein n is 0-2; each AA is independently L-valine or L-alanine; R₁ is selected from the group consisting of N-benzyloxycarbonyl and other groups; R₈, R₉, R₁₀ are each independently hydrogen, lower alkyl and other groups.

Mjalli *et al.* (*Bioorg. Med. Chem. Lett.*, 3, 2689-2692, **1993**) report the preparation of peptide phenylalkyl ketones as reversible inhibitors of ICE, such as:

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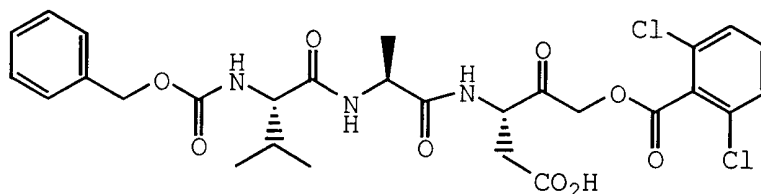


Thornberry *et al.* (*Biochemistry*, 33, 3934-3940, 1994) report the irreversible inactivation of ICE by peptide acyloxymethyl ketones:

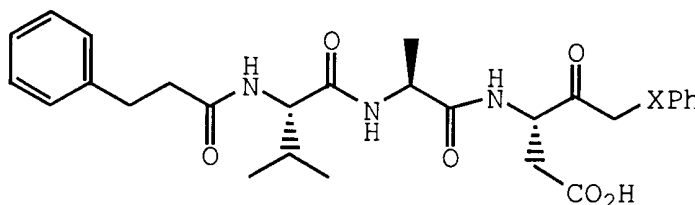


5 wherein Ar is C₆H₄-2,6-(CF₃)₂, C₆H₄-2,6-(CH₃)₂, Ph-F₅, and other groups.

Dolle *et al.* (*J. Med. Chem.* 37, 563-564, 1994) report the preparation of P₁ aspartate-based peptide α -((2,6-dichlorobenzoyl)oxy)methyl ketones as potent time-dependent inhibitors of ICE, such as:



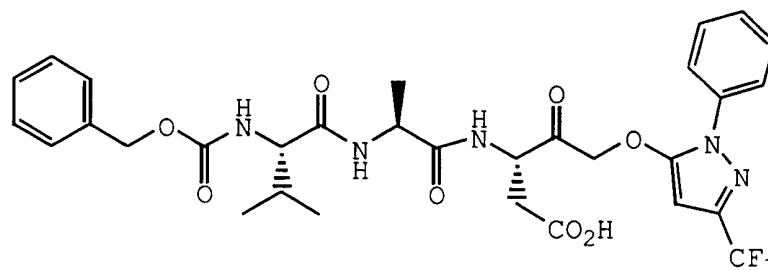
10 Mjalli *et al.* (*Bioorg. Med. Chem. Lett.*, 4, 1965-1968, 1994) report the preparation of activated ketones as potent reversible inhibitors of ICE:



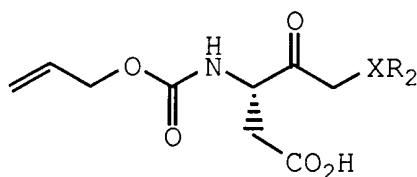
wherein X is NH(CH₂)₂, OCO(CH₂)₂, S(CH₂)₃, and other groups.

15 Dolle *et al.* (*J. Med. Chem.* 37, 3863-3866, 1994) report the preparation of α -((1-phenyl-3-(trifluoromethyl)-pyrazol-5-yl)oxy)methyl ketones as irreversible inhibitor of ICE, such as:

-6-

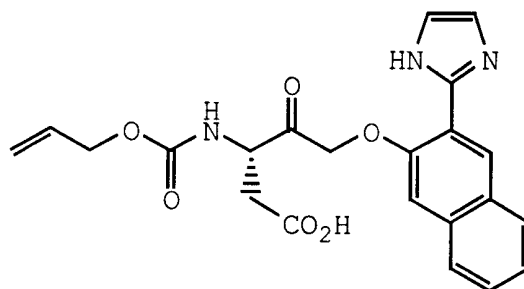


Mjalli *et al.* (*Bioorg. Med. Chem. Lett.*, 5, 1405-1408, **1995**) report inhibition of ICE by N-acyl-Aspartic acid ketones:

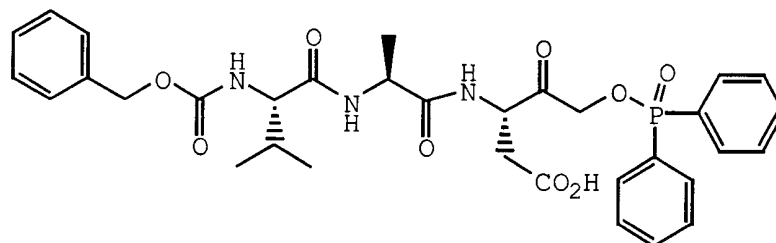


5 wherein XR_2 is $\text{NH}(\text{CH}_2)_2\text{Ph}$, $\text{OCO}(\text{CH}_2)_2\text{cyclohexyl}$ and other groups.

Mjalli *et al.* (*Bioorg. Med. Chem. Lett.*, 5, 1409-1414, **1995**) report inhibition of ICE by N-acyl-aspartyl aryloxymethyl ketones, such as:

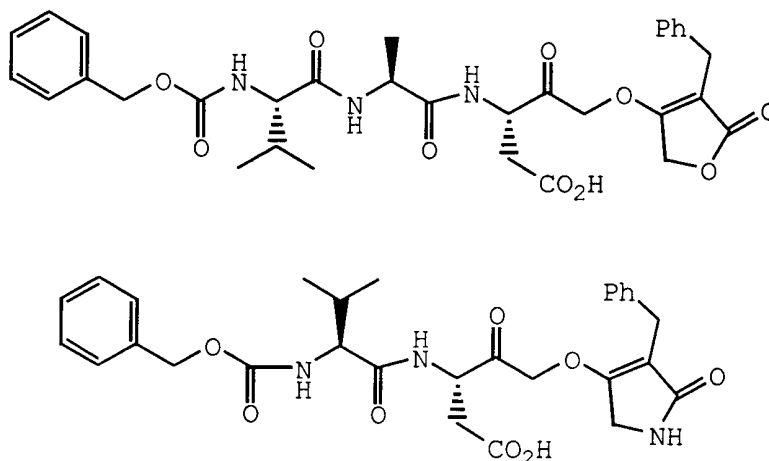


10 Dolle *et al.* (*J. Med. Chem.* 38, 220-222, **1995**) report the preparation of aspartyl α -((diphenylphosphinyl)oxy)methyl ketones as irreversible inhibitors of ICE, such as:

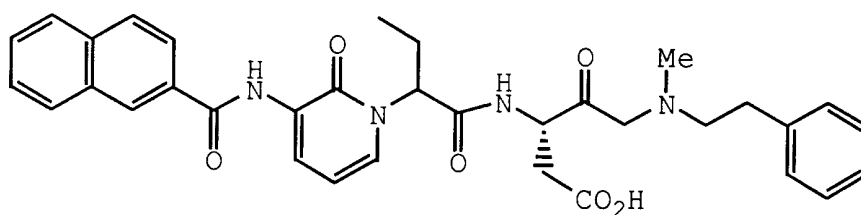


15 Graybill *et al.* (*Bioorg. Med. Chem. Lett.*, 7, 41-46, **1997**) report the preparation of α -((tetronoyl)oxy)- and α -((tetramoyl)oxy)methyl ketones as inhibitors of ICE, such as:

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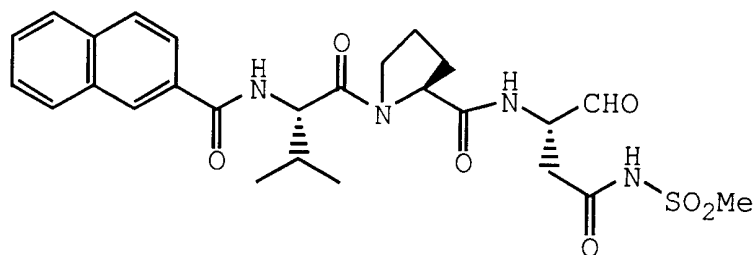


Semple *et al.* (*Bioorg. Med. Chem. Lett.*, 8, 959-964, **1998**) report the preparation of peptidomimetic aminomethylene ketones as inhibitors of ICE, such as:



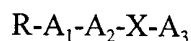
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Okamoto *et al.* (*Chem. Pharm. Bull.* 47, 11-21, **1999**) report the preparation of peptide based ICE inhibitors with the P1 carboxyl group converted to an amide, such as:

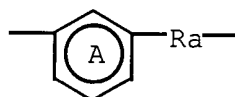


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EP618223 patent application disclosed inhibitor of ICE as anti-inflammatory agents:



Wherein R is a protecting group or optionally substituted benzyloxy; A_1 is an α -hydroxy or α -amino acid residue or a radical of formula:

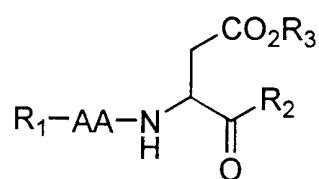


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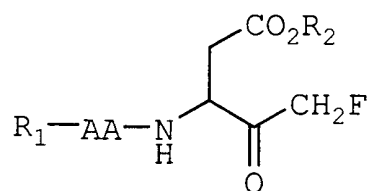
wherein ring A is optionally substituted by hydroxy or C₁₋₄ alkoxy and R_a is CO or CS; A₂ is an α-hydroxy or α-amino acid residue or A₁ and A₂ form together a pseudo-dipeptide or a dipeptide mimetic residue; X is a residue derived from Asp; A₃ is -CH₂-X₁-CO-Y₁, -CH₂-O-Y₂, -CH₂-S-Y₃, wherein X₁ is O or S; Y₁, Y₂ or Y₃ is cycloaliphatic residue, and optionally substituted aryl.

WO99/18781 discloses dipeptides of formula:



wherein R₁ is an N-terminal protecting group; AA is a residue of any natural or non-natural α-amino acid, β-amino acid, derivatives of an α-amino acid or β-amino acid; R₂ is H or CH₂R₄ where R₄ is an electronegative leaving group, and R₃ is alkyl or H, provided that AA is not His, Tyr, Pro or Phe. These dipeptides are surprisingly potent caspase inhibitors of apoptosis in cell based systems. These compounds are systemically active *in vivo* and are potent inhibitors of antiFas-induced lethality in a mouse liver apoptosis model and have robust neuroprotective effects in a rat model of ischemic stroke.

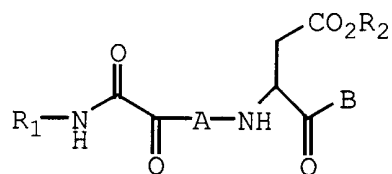
WO 99/47154 discloses dipeptides of formula:



wherein R₁ is an N-terminal protecting group; AA is a residue of a non-natural α-amino acid or β-amino acid; R₂ is an optionally substituted alkyl or H.

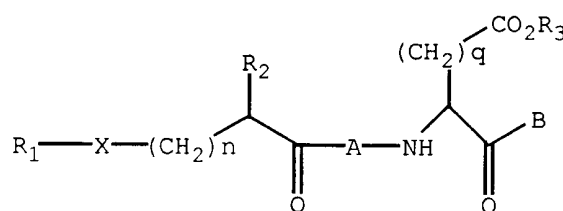
WO 00/01666 discloses c-terminal modified oxamyl dipeptides as inhibitors of the ICE/ced-3 family of cysteine proteases:

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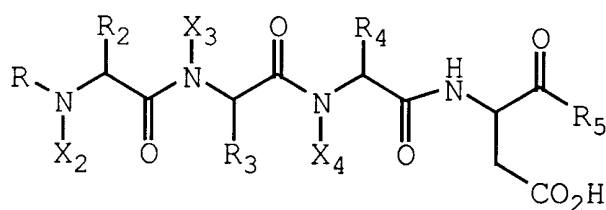
wherein A is a natural or unnatural amino acid; B is a hydrogen atom, a deuterium atom, alkyl, cycloalkyl and other groups; R₁ is alkyl, cycloalkyl and other groups, R₂ is hydrogen, lower alkyl and other groups.

5 WO 00/23421 discloses (substituted)acyl dipeptidyl inhibitors of the ICE/ced-3 family of cysteine proteases:



wherein A is a natural or unnatural amino acid; B is a hydrogen atom, a deuterium atom, alkyl, cycloalkyl and other groups; X is CH₂, C=O, O, S, NH, C=ONH or CH₂OC=ONH; n is 0, 1, or 2; q is 1 or 2; R₁ is optionally substituted aryl and heteroaryl; R₂ is hydrogen, alkyl and other groups; R₃ is hydrogen, alkyl and other groups.

WO 00/32620 discloses gamma-ketoacid tetrapeptides as inhibitors of the caspase-3:



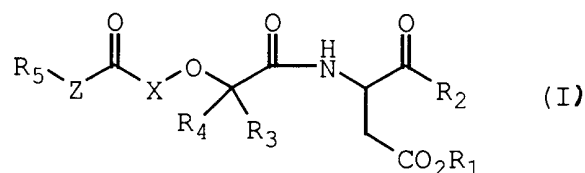
15 wherein R is H and COR₁; R₁ is hydrogen, alkoxy and other groups; R₂ is hydrogen, CH₃ and other groups; R₃ is hydrogen, CH₃ and other groups; R₄ is hydrogen, CH₃ and other groups; R₅ is C₁₋₆alkyl, arylC₁₋₆alkyl and other groups; X₂, X₃ and X₄ are independently H or X₂ and R₂, X₃ and R₃ or X₄ and R₄ may together form a saturated monocyclic ring.

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Summary of the Invention

The invention relates to compound of Formula I:



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or pharmaceutically acceptable salts or prodrugs thereof, wherein

R₁ is an optionally substituted alkyl or hydrogen;

R₂ is hydrogen or optionally substituted alkyl;

10 R₃ and R₄ independently are hydrogen, optionally substituted aryl, optionally substituted heterocyclic, optionally substituted carbocyclic, optionally substituted heteroaryl, optionally substituted alkyl, optionally substituted alkenyl, or optionally substituted alkynyl;

15 R₅ is optionally substituted alkyl, optionally substituted carbocyclic, optionally substituted heterocyclic, optionally substituted aryl or optionally substituted heteroaryl;

Z is O, S, NR₈, or (CR₉R₁₀)_n, where R₈, R₉ and R₁₀ independently are hydrogen, alkyl or cycloalkyl, and n is 0, 1, 2, or 3; and

X is a peptide of 1-2 amino acids or a bond.

20 The present invention provides pharmaceutical compositions comprising a compound of Formula I in an effective amount to reduce apoptotic cell death in an animal.

25 The invention relates to the discovery that the compounds represented by Formula I are inhibitors of caspases. The invention also relates to the use of the compounds of the invention for reducing, preventing or treating maladies in which apoptotic cell death is either a causative factor or a result. Examples of uses for the present invention include treating or ameliorating cell death in the central or peripheral nervous system, retinal neurons, cardiac muscle or immune system cells of an animal; treating or preventing polycystic

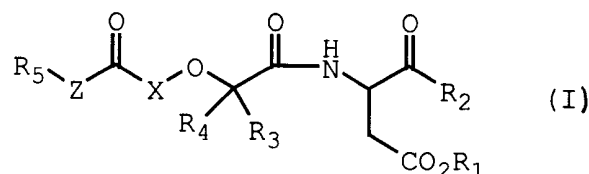
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kidney disease, renal amyloidosis, acute renal failure, cyclosporine A induced murin tubular epithelial cell death, HIV-induced nephropathy or anemia/erythropoiesis in an animal; protecting a mammalian organ or tissue from cell death due to deprivation of normal blood supply; reducing or preventing cell death in a donor organ or tissue after it has been transplanted into a host due to the effects of host immune cells; reducing or preventing the death of mammalian sperm or eggs used in *in vitro* fertilization procedures; extending the lifespan of a mammalian or yeast cell line; treating or ameliorating hair loss or premature graying of the hair in a mammal; treating or ameliorating skin damage of a mammal due to exposure to high levels of radiation, heat or chemicals; treating or ameliorating sepsis in an animal; treating or ameliorating hepatitis in an animal; treating or ameliorating hereditary tyrosinemia type 1 in an animal; treating or ameliorating chronic alcohol ingestion induced buccal mucosa cell death in an animal; treating or ameliorating cell death in plants or flowers; treating or ameliorating radiation or ultraviolet-irradiation induced cell death in an animal; treating or ameliorating apoptotic death of bone marrow cells in myelodysplastic syndromes (MDS); treating or ameliorating apoptotic cell death in acute pancreatitis; treating or preventing the inflammatory response in psoriasis or inflammatory bowel disease; treating or ameliorating organ apoptosis after burn injury; treating or ameliorating small bowel tissue injury after intestinal ischemia-reperfusion; and treating, ameliorating or preventing oral mucositis, gastrointestinal mucositis, bladder mucositis, proctitis, bone marrow cell death, skin cell death or hair loss resulting from chemotherapy or radiation therapy of cancer in an animal

Detailed Description of the Invention

The inhibitors of caspases and apoptotic cell death of the present invention are compounds having the general Formula I:

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or pharmaceutically acceptable salts or prodrugs thereof, wherein

5 R₁ is an optionally substituted alkyl or hydrogen;

R₂ is hydrogen or optionally substituted alkyl;

10 R₃ and R₄ independently are hydrogen, optionally substituted aryl, optionally substituted heterocyclic, optionally substituted carbocyclic, optionally substituted heteroaryl, optionally substituted alkyl, optionally substituted alkenyl, or optionally substituted alkynyl;

R₅ is an optionally substituted alkyl, optionally substituted carbocyclic, optionally substituted heterocyclic, optionally substituted aryl or optionally substituted heteroaryl;

15 Z is O, S, NR₈, or (CR₉R₁₀)_n, where R₈, R₉ and R₁₀ independently are hydrogen, alkyl or cycloalkyl, and n is 0, 1, 2, or 3; and

20 X is a peptide of 1-2 amino acids or a bond. Where X is one amino acid, it may be any one of the common 20 amino acids e.g., Ala, Val, Leu, Ile, Pro, Phe, Trp, Met, Gly, Ser, Thr, Cys, Tyr, Asp, Asn, Glu, Asn, Lys, Arg and His. Where X is a peptide, it may be Asp-Glu, Asp-Ala, Asp-Phe, Val-Glu, Leu-Glu, Thr-Glu, Ile-Glu, Tyr-Glu, and Trp-Glu.

With respect to R₁, preferred alkyl groups are C₁₋₆ alkyl groups, e.g. methyl, ethyl, propyl, isopropyl, isobutyl, pentyl and hexyl groups; and substituted C₁₋₆ alkyl groups, e.g. CH₂OCH₃ and CH₂OCOCH₃ (AM).

25 Preferred R₂ are alkyl group substituted by electronegative group or leaving group, including fluoromethyl, chloromethyl, alkoxymethyl, aryloxymethyl, heteroaryloxymethyl, alkylthiomethyl, arylthiomethyl, aminomethyl, acyloxymethyl, and arylacyloxymethyl. Other examples of optional substituents that may be present at the R₂ alkyl group include, without limitation, 3-pyrazolyloxy optionally substituted at the 2, 4 and 5-positions
30 with lower alkyl; 3-(1-phenyl-3-trifluoromethyl)pyrazolyloxy; 2,6-

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di(trifluoromethyl)benzoyloxy; 2,6-dimethylbenzoyloxy, pentafluorophenoxy; 2,6-dichlorobenzoyloxy; 2-(3-(2-imidazolyl)naphthyl)oxy; diphenylphosphinyloxy; tetronyloxy; and tetramoyloxy.

5 With respect to R_3 and R_4 , preferred groups are those that will match the side chain of natural amino acids or non-natural amino acids. In one embodiment, preferred R_3 is hydrogen, preferred R_4 is straight-chained or branched C_{1-6} alkyl including methyl, ethyl, isopropyl and isobutyl; or preferred R_4 is cycloalkyl, aryl or heteroaryl; or preferred R_4 is straight-chained or branched C_{1-6} alkyl optionally substituted by one or more hydroxy,
10 halogen, carboxy, amino, amide, ester, guanadino, thiol, alkylthiol, aryl, heterocyclic or heteroaryl groups.

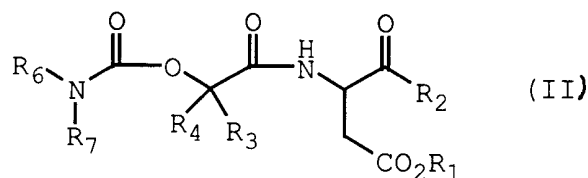
With respect to R_5 , preferred alkyl are methyl, ethyl, isopropyl, isobutyl; preferred substituents on alkyl are hydroxy, carboxy, halogen, C_4 - C_7 cycloalkyl, saturated and unsaturated heterocyclic, aryl or heteroaryl; preferred
15 cycloalkyl are cyclopentyl and cyclohexyl; preferred saturated and unsaturated heterocyclic groups are piperidinyl and morpholinyl; preferred aryls are phenyl and naphthyl; preferred heteroaryls are pyridyl, indolyl, furyl and thienyl; preferred substituents on the aryl and heteroaryl are methyl, ethyl, chloro, fluoro, bromo, trifluoromethyl, methoxy, hydroxy, carboxy, cyano and nitro.
20 Especially preferred R_5 is optionally substituted phenyl, naphthyl, heteroaryl, or benzyl. Preferred Z is O, NH, or CH_2 .

The invention relates to the discovery that the compounds represented by Formula I are inhibitors of caspases. These inhibitors slow or block cell death in a variety of clinical conditions and industrial applications in which the
25 loss of cells, tissues or entire organs occurs. Therefore, the invention is also related to methods of treating, preventing or reducing conditions in which apoptosis plays a role. These conditions are more fully described below.

The methods comprise administering to an animal in need of such treatment an inhibitor of the present invention, or a pharmaceutically
30 acceptable salt or prodrug thereof, in an amount effective to inhibit apoptotic cell death.

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Preferred embodiments of the compounds of Formula I that may be employed as inhibitors of caspases are represented by Formula II:



or pharmaceutically acceptable salts or prodrugs thereof wherein R_1 - R_4 are as defined previously with respect to Formula I. R_6 and R_7 independently are hydrogen, alkyl, optionally substituted alkyl, C_4 - C_7 cycloalkyl, saturated or unsaturated heterocyclic, aryl, heteroaryl, or R_6 and R_7 are combined together with the nitrogen to form a heterocycle.

Preferred R_1 is H, Me, Et, t-Bu or AM. Preferred R_2 is fluoromethyl, acyloxymethyl, arylacyloxymethyl, aryloxymethyl, heteroaryloxymethyl and aminomethyl. With respect to R_6 and R_7 , preferred alkyl are methyl, ethyl, isopropyl, isobutyl; preferred substituents on alkyl are hydroxy, carboxy, carbamyl, carboxy ester, halogen, C_4 - C_7 cycloalkyl, saturated and unsaturated heterocyclic, aryl or heteroaryl; preferred cycloalkyl are cyclopentyl and cyclohexyl; preferred saturated and unsaturated heterocyclic groups are piperidiny and morpholinyl; preferred aryls are phenyl and naphthyl; preferred heteroaryls are pyridyl, indolyl, furyl and thienyl; preferred substituents on the aryl and heteroaryl are methyl, ethyl, chloro, fluoro, bromo, trifluoromethyl, methoxy, hydroxy, carboxy, cyano and nitro. Preferred heterocycles from combination of R_6 and R_7 together with the nitrogen are piperidine, morpholine, and piperazine.

With respect to R_3 and R_4 , preferred R_3 is hydrogen, and preferred R_4 are those that will match the side chain of natural amino acids or non-natural amino acids. Specifically, preferred R_3 and R_4 , together with the attached carbon, oxygen and carbonyl group are a residue of an α -hydroxy acid. These α -hydroxy acids include, but are not limited to 3-methoxymandelic acid, 4-methoxymandelic acid, 2-methoxymandelic acid, 4-bromomandelic acid, 4-methylmandelic acid, 2,5-dimethylmandelic acid, 4-methylthiomandelic acid,

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4-chloromandelic acid, 3-chloromandelic acid, 2-chloromandelic acid, 4-fluoromandelic acid, 2,3-difluoromandelic acid, 2,4-difluoromandelic acid, 2,5-difluoromandelic acid, 3,5-difluoromandelic acid, 3,4-difluoromandelic acid, 2,6-difluoromandelic acid, 2,3,5-trifluoromandelic acid, 2,3,6-trifluoromandelic acid, 3,4-methylenedioxymandelic acid, 4-trifluoromethylmandelic acid, 3-nitromandelic acid, 4-nitromandelic acid, mandelic acid, hexahydromandelic acid, 2-hydroxy-3-methylbutyric acid, 2-hydroxy-3,3-dimethylbutyric acid, 2-hydroxy-4-phenylbutyric acid, 2-hydroxybutyric acid, 2-hydroxybutynoic acid, 2-hydroxy-4-(methylthio)butyric acid, 2-hydroxy-3-butenic acid, 2-hydroxy-3-methylpentanoic acid, 2-hydroxy-4-methylpentanoic acid, 3-phenyllactic acid, lactic acid, 3-chlorolactic acid, 3-imidazolelactic acid, 3,3,3-trifluorolactic acid, 3,3-difluoro-3-(4-fluorophenyl)lactic acid, 3,3-difluoro-3-(4-methoxyphenyl)lactic acid, 3,3-difluoro-3-(4-chlorophenyl)lactic acid, 3,3-difluoro-3-phenyllactic acid, 3-(1-naphthoxy)lactic acid, leucic acid, glycolic acid, 2-hydroxycaproic acid, 2-hydroxyvaleric acid, 2-hydroxyoctanoic acid, and indole-3-lactic acid. These α -hydroxy acids are commercially available from Aldrich, Sigma, Lancaster, Maybridge and other companies.

Exemplary preferred inhibitors of caspases having Formulas I and II include, without limitation:

1-(Carbonyl-Asp-CH₂F)ethyl N-phenylcarbamate,
 1-(Carbonyl-Asp-CH₂F)ethyl N-benzylcarbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-phenylcarbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-benzylcarbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-(2,6-dichlorophenyl)carbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-(2,5-dichlorophenyl)-carbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-(2,4-dichlorophenyl)-carbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂DCB)propyl N-phenylcarbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂DCB)propyl N-(2,6-dichlorophenyl)-carbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂PTP)propyl N-phenylcarbamate,

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- 2-Methyl-1-(carbonyl-Asp-CH₂PTP)propyl *N*-(2,6-dichlorophenyl)-carbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂DPP)propyl *N*-phenylcarbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂DPP)propyl *N*-(2,6-dichlorophenyl)-carbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂F)propyl *N*-(2-methyl-1-methoxycarbonyl-
 5 propyl)carbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂F)propyl *N*-(3-fluorophenyl)carbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂F)propyl *N*-(4-fluorophenyl)carbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂F)propyl *N*-(3,4-difluorophenyl)carbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂F)propyl *N*-(4-phenoxyphenyl)carbamate,
 10 1-(Carbonyl-Asp-CH₂F)propyl *N*-phenylcarbamate,
 1-(Carbonyl-Asp-CH₂F)butyl *N*-phenylcarbamate,
 1-(Carbonyl-Asp-CH₂F)-2-propenyl *N*-phenylcarbamate,
 2-(4-Imidazolyl)-1-(carbonyl-Asp-CH₂F)ethyl *N*-phenylcarbamate,
 2-Phenyl-1-(carbonyl-Asp-CH₂F)ethyl *N*-phenylcarbamate,
 15 2-Methyl-1-(carbonyl-Asp-CH₂F)butyl *N*-phenylcarbamate,
 3-Methyl-1-(carbonyl-Asp-CH₂F)butyl *N*-phenylcarbamate,
 1-Phenyl-1-(carbonyl-Asp-CH₂F)methyl *N*-phenylcarbamate,
 1-(2-Chlorophenyl)-1-(carbonyl-Asp-CH₂F)methyl *N*-phenylcarbamate,
 1-(4-Chlorophenyl)-1-(carbonyl-Asp-CH₂F)methyl *N*-phenylcarbamate,
 20 1-Cyclohexyl-1-(carbonyl-Asp-CH₂F)methyl *N*-phenylcarbamate,
 2-Chloro-1-(carbonyl-Asp-CH₂F)ethyl *N*-phenylcarbamate,
 2,2,2-Trifluoro-1-(carbonyl-Asp-CH₂F)ethyl *N*-phenylcarbamate, and
Z-Valine 2-methyl-1-(carbonyl-Asp-CH₂F)propyl ester.
 where *Z* is benzyloxycarbonyl, DPP is diphenylphosphinyloxy, PTP is 1-
 25 phenyl-3-(trifluoromethyl)pyrazol-5-yloxy and DCB is 2,6-
 dichlorobenzoyloxy.

Useful aryl groups are C₆₋₁₄ aryl, especially C₆₋₁₀ aryl. Typical C₆₋₁₄ aryl groups include phenyl, naphthyl, phenanthrenyl, anthracenyl, indenyl, azulenyl, biphenyl, biphenylenyl and fluorenyl groups.

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Useful cycloalkyl groups are C₃₋₈ cycloalkyl. Typical cycloalkyl groups include cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl and cycloheptyl.

5 Useful saturated or partially saturated carbocyclic groups are cycloalkyl groups as defined above, as well as cycloalkenyl groups, such as cyclopentenyl, cycloheptenyl and cyclooctenyl.

Useful halo or halogen groups include fluorine, chlorine, bromine and iodine.

10 Useful alkyl groups include straight-chained and branched C₁₋₁₀ alkyl groups, more preferably C₁₋₆ alkyl groups. Typical C₁₋₁₀ alkyl groups include methyl, ethyl, propyl, isopropyl, butyl, *sec*-butyl, *tert*-butyl, 3-pentyl, hexyl and octyl groups. Also contemplated is a trimethylene group substituted on two adjoining positions on the benzene ring of the compounds of the invention.

15 Useful alkenyl groups include straight chained and branched C₂₋₁₀ alkenyl groups, more preferably C₂₋₆ alkenyl groups. Typical C₂₋₁₀ alkenyl groups include ethenyl, propenyl, isopropenyl, butenyl, pentenyl, hexenyl and octenyl groups.

20 Useful alkynyl groups include straight chained and branched C₂₋₁₀ alkynyl groups, more preferably C₂₋₆ alkynyl groups. Typical C₂₋₁₀ alkynyl groups include ethynyl, propynyl, butynyl, pentynyl, hexynyl and octynyl groups.

25 Useful arylalkyl groups include any of the above-mentioned C₁₋₁₀ alkyl groups substituted by any of the above-mentioned C₆₋₁₄ aryl groups. Useful values include benzyl, phenethyl and naphthylmethyl.

Useful haloalkyl groups include C₁₋₁₀ alkyl groups substituted by one or more fluorine, chlorine, bromine or iodine atoms, e.g. fluoromethyl, difluoromethyl, trifluoromethyl, pentafluoroethyl, 1,1-difluoroethyl, chloromethyl, chlorofluoromethyl and trichloromethyl groups.

30 Useful alkoxy groups include oxygen substituted by one of the C₁₋₁₀ alkyl groups mentioned above.

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Useful alkylthio groups include sulphur substituted by one of the C₁₋₁₀ alkyl groups mentioned above. Also included are the sulfoxides and sulfones of such alkylthio groups.

5 Useful acylamino groups are any C₁₋₆ acyl (alkanoyl) attached to an amino nitrogen, e.g. acetamido, propionamido, butanoylamido, pentanoylamido, hexanoylamido as well as aryl-substituted C₂₋₆ substituted acyl groups.

10 Useful acyloxy groups are any C₁₋₆ acyl (alkanoyl) attached to an oxy (-O-) group, e.g. formyloxy, acetoxy, propionyloxy, butanoyloxy, pentanoyloxy, hexanoyloxy and the like.

Useful arylacyloxy groups include any of the aryl groups mentioned above substituted on any of the acyloxy groups mentioned above, e.g. 2,6-dichlorobenzoyloxy, 2,6-difluorobenzoyloxy and 2,6-di-(trifluoromethyl)-benzoyloxy groups.

15 Useful amino groups include -NH₂, -NHR₁₁, and -NR₁₁R₁₂, wherein R₁₁ and R₁₂ are C₁₋₁₀ alkyl or cycloalkyl groups as defined above.

20 Useful saturated or partially saturated heterocyclic groups include tetrahydrofuranyl, pyranal, piperidinyl, piperizinyl, pyrrolidinyl, imidazolidinyl, imidazoliny, indoliny, isoindoliny, quinuclidiny, morpholiny, isochromanyl, chromanyl, pyrazolidiny, pyrazoliny, tetronoyl and tetramoyl groups.

25 Useful heteroaryl groups include any one of the following: thienyl, benzo[b]thienyl, naphtho[2,3-b]thienyl, thianthrenyl, furyl, pyranal, isobenzofuranyl, chromenyl, xanthenyl, phenoxanthiiny, 2H-pyrrolyl, pyrrolyl, imidazolyl, pyrazolyl, pyridyl, pyraziny, pyrimidinyl, pyridaziny, indoliziny, isoindolyl, 3H-indolyl, indolyl, indazolyl, puriny, 4H-quinoliziny, isoquinolyl, quinolyl, phthalziny, naphthyridiny, quinozaliny, cinnoliny, pteridinyl, carbazolyl, β-carboliny, phenanthridiny, acridiny, perimidiny, phenanthroliny, phenaziny, isothiazolyl, phenothiaziny, isoxazolyl, furazany, phenoxaziny, 30 1,4-dihydroquinoxaline-2,3-dione, 7-aminoisocoumarin, pyrido[1,2-

a]pyrimidin-4-one, 1,2-benzisoxazol-3-yl, benzimidazolyl, 2-oxindolyl and 2-oxobenzimidazolyl. Where the heteroaryl group contains a nitrogen atom in a ring, such nitrogen atom may be in the form of an N-oxide, e.g. a pyridyl N-oxide, pyrazinyl N-oxide, pyrimidinyl N-oxide and the like.

5 Optional substituents include one or more alkyl; halo; haloalkyl; cycloalkyl; aryl optionally substituted with one or more lower alkyl, halo, haloalkyl or heteroaryl groups; aryloxy optionally substituted with one or more lower alkyl, halo, haloalkyl or heteroaryl groups; aralkyl; heteroaryl optionally substituted with one or more lower alkyl, haloalkyl and aryl groups;
10 heteroaryloxy optionally substituted with one or more lower alkyl, haloalkyl and aryl groups; alkoxy; alkylthio; arylthio; amino; acyloxy; arylacyloxy optionally substituted with one or more lower alkyl, halo alkyl and aryl groups; diphenylphosphinyloxy optionally substituted with one or more lower alkyl, halo or haloalkyl groups; heterocyclo optionally substituted with one or more
15 lower alkyl, haloalkyl and aryl groups; heterocycloalkyloxy optionally substituted with one or more lower alkyl, haloalkyl and aryl groups; partially unsaturated heterocycloalkyl optionally substituted with one or more lower alkyl, haloalkyl and aryl groups; or partially unsaturated heterocycloalkyloxy optionally substituted with one or more lower alkyl, haloalkyl and aryl groups.

20 Certain of the compounds of the present invention may exist as stereoisomers including optical isomers. The invention includes all stereoisomers and both the racemic mixtures of such stereoisomers as well as the individual enantiomers that may be separated according to methods that are well known to those of ordinary skill in the art.

25 Examples of pharmaceutically acceptable addition salts include inorganic and organic acid addition salts such as hydrochloride, hydrobromide, phosphate, sulphate, citrate, lactate, tartrate, maleate, fumarate, mandelate and oxalate; and inorganic and organic base addition salts with bases such as sodium hydroxy and Tris(hydroxymethyl)aminomethane (TRIS, tromethane).

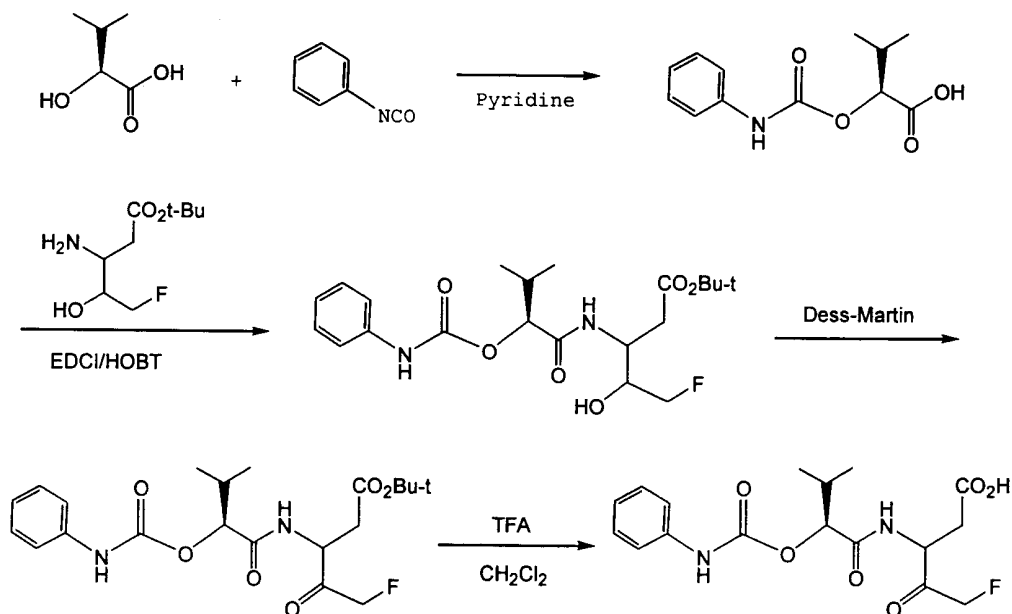
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Examples of prodrugs include compounds of Formulae I-II wherein R₁ is an alkyl group or substituted alkyl group such as CH₂OCH₃ and CH₂OCOCH₃ (AM ester).

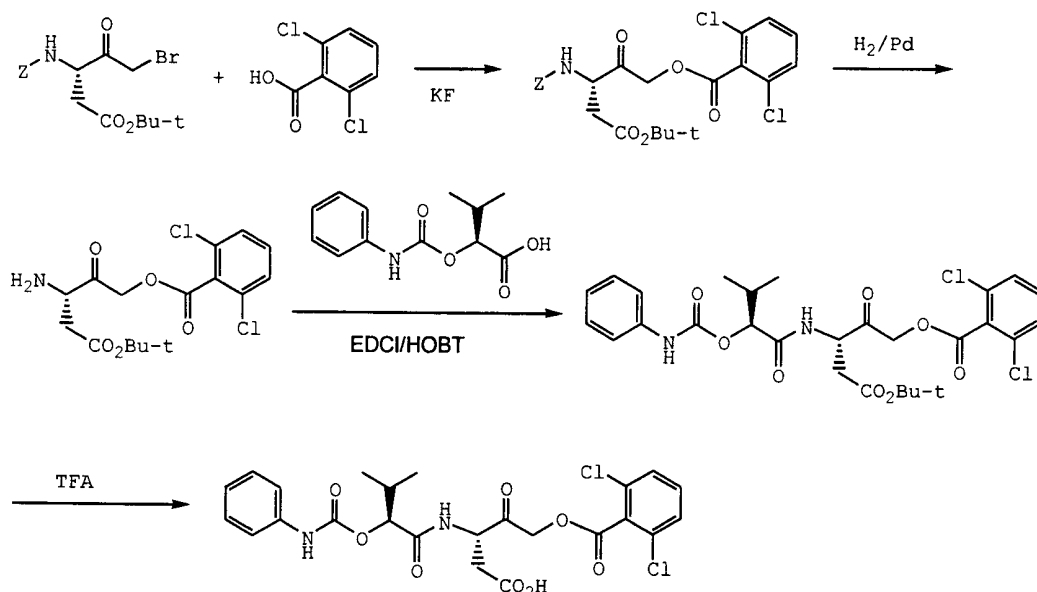
5 The invention is also directed to a method for treating disorders responsive to the inhibition of caspases in animals suffering thereof. Particular preferred embodiments of compounds for use in the method of this invention are represented by previously defined Formulae I-II.

10 The compounds of this invention may be prepared using methods known to those skilled in the art. Specifically, compounds of Formulae I-II with a fluoromethyl (CH₂F) group can be prepared as illustrated by exemplary reactions in Scheme 1. Reaction of hydroxy acid with phenyl isocyanate gave the carbamate protected acid. Coupling of the acid with the intermediate hydroxy amine (Revesz *et al.*, *Tetrahedron Lett.* 35:9693-9696 (1994)) produced the amide. Oxidation of the hydroxy group by Dess-Martin reagent
15 according to Revesz *et al.* (*Tetrahedron Lett.* 35:9693-9696 (1994)) gave the ketone as a mixture of diastereomers. Other oxidation reagents can be used in place of Dess-Martin reagent include pyridinium chlorochromate (PCC). TFA catalyzed cleavage of the *t*-Bu ester produced the acid.

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Scheme 1

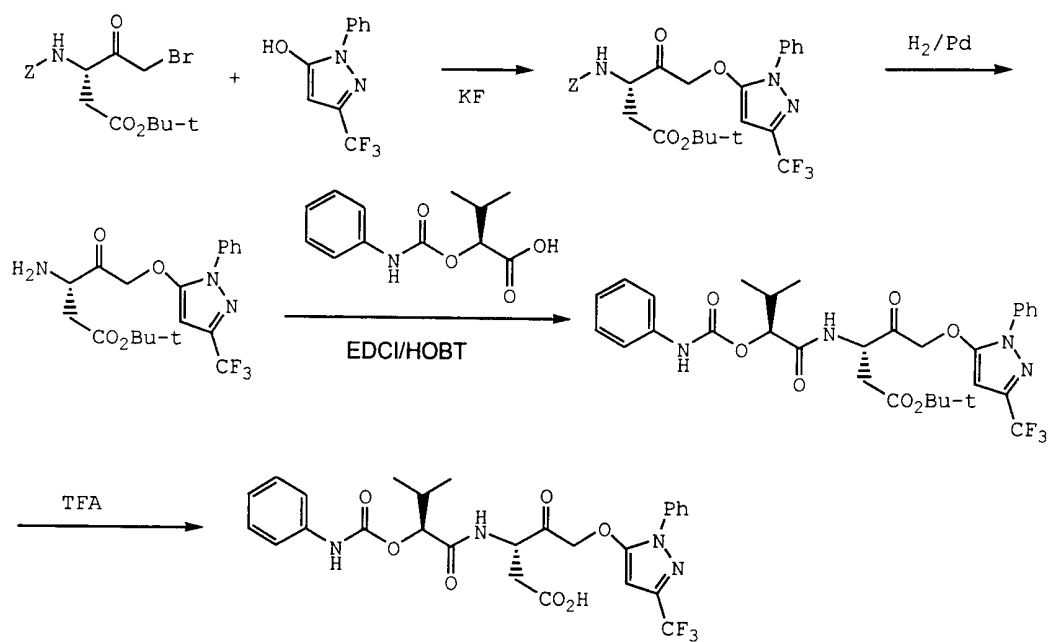
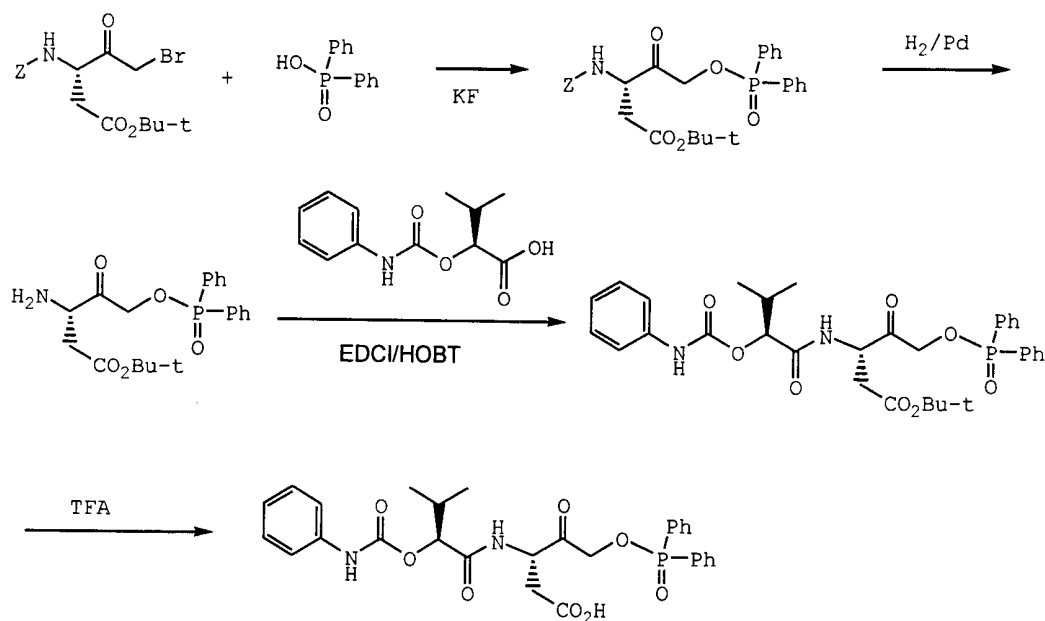
- 5 Compounds of Formulae I-II with a dichlorobenzoyloxy (DCB) group can be prepared as illustrated by exemplary reactions in Scheme 2. Reaction of the bromomethylketone with dichlorobenzoic acid gave the ester. The Z protecting group was removed by hydrogenation to give the amine, which was coupled with the acid to produce the amide. The t-Bu ester was cleaved by
- 10 TFA to give the acid.

Scheme 2

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Similarly, compounds of Formulae I-II with a 1-phenyl-3-(trifluoromethyl)pyrazol-5-yloxy (PTP) and a diphenylphosphinyloxy (DPP) group can be prepared as illustrated by exemplary reactions in Scheme 3 and 4, respectively.

5

Scheme 3**Scheme 4**

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An important aspect of the present invention is the discovery that compounds having Formulae I-II are inhibitors of caspases. Therefore, these inhibitors are expected to slow or block cell death in a variety of clinical conditions in which the loss of cells, tissues or entire organs occurs.

5 The cell death inhibitors of the present invention can be used to reduce or prevent cell death in the nervous system (brain, spinal cord, and peripheral nervous system) under various conditions of ischemia and excitotoxicity, including, but not limited to, focal ischemia due to stroke and global ischemia due to cardiac arrest, as well as spinal cord injury (Emery *et al.*, *J. Neurosurgery* 89:911-920 (1998) and Springer *et al.*, *Nat. Med.* 5:943-946 (1999)). One particular usage is to treat the effects of oxygen deprivation which can occur during the birth of infants in high-risk labors or drowning. The cell death inhibitors can also be used to reduce or prevent cell death in the nervous system due to traumatic injury (such as head trauma), viral infection or radiation-induced nerve cell death (for example, as a side-effect of cancer radiotherapy). The cell death inhibitors can also be used to reduce or prevent cell death in a range of neurodegenerative disorders, including but not limited to Alzheimer's disease (Mattson *et al.*, *Brain Res.* 807:167-176 (1998)), Huntington's Disease, Parkinson's Disease, a prion disease, multiple sclerosis, 15 amyotrophic lateral sclerosis, and spinobulbar atrophy. The *in vivo* neuroprotective properties of cell death inhibitors of the invention can be tested in a rat transient focal brain ischemia model (Xue *et al.*, *Stroke* 21:166 (1990)).

20 The cell death inhibitors can also be used to treat or ameliorate cell death in the central or peripheral nervous system due to acute bacterial meningitis.

25 The cell death inhibitors of the invention can be used to reduce or prevent cell death in any conditions which potentially results in the death of cardiac muscle (Black *et al.*, *J. Mol. Cel. Card.* 30:733-742 (1998) and Maulik *et al.*, *Free Radic. Biol. Med.* 24:869-875 (1998)). This includes myocardial infarction due to myocardial ischemia and reperfusion, congestive heart failure 30

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and cardiomyopathy. One particular application is to reduce or prevent myocardial cell death as occurs in certain viral infections of the heart.

The *in vivo* activity of the cell death inhibitors of the invention can be tested using the "mouse liver apoptosis" model described by Rodriguez et al. (Rodriguez *et al.*, *J. Exp. Med.* 184:2067-2072 (1996)). In this model, mice are treated intravenously (IV) with an antiFas antibody which induces massive apoptosis in the liver and other organs, leading to generalized organ failure and death. This model is useful for indirectly testing the systemic bioavailability of the cell death inhibitors of the invention, as well as their *in vivo* anti-apoptotic properties. The cell death inhibitors of the invention therefore can be used to reduce or prevent apoptosis of liver cells (Jones *et al.*, *Hepatology* 27:1632-1642 (1998)) such as in sepsis (Jaeschke *et al.*, *J. Immunol.* 160:3480-3486 (1998)) and hereditary tyrosinemia type 1 (HT1) (Kubo *et al.*, *Proc. Natl. Acad. Sci. USA* 95:9552-9557 (1998)). The cell death inhibitors of the invention also can be used to treat hepatitis (Suzuki, *Proc. Soc. Exp. Biol. Med.* 217:450-454 (1998)).

The cell death inhibitors of the invention can be used to reduce or prevent cell death of retinal neurons (Kermer *et al.*, *J. Neurosci.* 18:4656-4662 (1998) and Miller *et al.*, *Am. J. Vet. Res.* 59:149-152 (1998)) as can occur in disorders which increase intraocular pressure (such as glaucoma) or retinal disorders associated with the aging process (such as age-related macular degeneration). The inhibitors can also be used to treat hereditary degenerative disorders of the retina, such as retinitis pigmentosa.

The cell death inhibitors of the invention can also be used to reduce or prevent cell death in the kidney. This includes renal amyloidosis (Hiraoka *et al.*, *Nippon Jinzo Gakkai Shi* 40:276-83 (1998)), acute renal failure (Lieberthal *et al. Semin Nephrol.* 18:505-518 (1998)), murine tubular epithelial cell death induced by cyclosporine A (Ortiz *et al. Kidney International Supp.* 68:S25-S29 (1998)) and HIV-induced nephropathy (Conaldi *et al. J. Clin. Invest.* 102:2041-2049 (1998)).

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The cell death inhibitors of the invention can also be used to reduce or prevent cell death of buccal mucosa due to chronic alcohol ingestion (Slomiany *et al.*, *Biochem. Mol. Biol. Int.* 45:1199-1209 (1998)).

5 The cell death inhibitors of the invention can also be used to reduce or prevent apoptotic cell death in acute pancreatitis (Iovanna, *Int. J. Pancreatol.* 20:77-84 (1996) and Sata *et al.*, *Pancreas* 19:76-82 (1999)).

The cell death inhibitors of the invention can also be used to reduce or prevent cell death in plants (Richberg *et al.*, *Curr. Opin. Plant Biol.* 1:480-485 (1998)), such as plant cell death due to pathogens (Pozo *et al.*, *Curr. Biol.* 10 8:1129-1132 (1998) and Greenberg *et al.*, *Cell*, 77:551-563 (1994)).

The cell death inhibitors of the invention can also be used to reduce or prevent cell death due to radiation and ultraviolet-irradiation (Sheikh *et al.*, *Oncogene* 17:2555-2563 (1998)).

15 The cell death inhibitors of the invention can also be used to reduce or prevent apoptotic death of bone marrow cells in myelodysplastic syndromes (MDS) (Mundle *et al.*, *Am. J. Hematol.* 60:36-47 (1999)).

The cell death inhibitors of the invention can also be used to reduce or prevent organ apoptosis, such in thymus and spleen, in the early period after burn injury (Fukuzuka *et al.*, *Ann. Surg.* 229:851-858 (1999)).

20 The cell death inhibitors of the invention can also be used to reduce or prevent premature death of cells of the immune system, and are particularly useful in treating immune deficiency disorders, such as acquired immune deficiency syndrome (AIDS), severe combined immune deficiency syndrome (SCIDS) and related diseases. The cell death inhibitors can also be used to
25 treat radiation-induced immune suppression.

The cell death inhibitors can also be used to treat or ameliorate an autoimmune disorder including lupus erythematosus, rheumatoid arthritis and type I diabetes.

30 Transplantation of human organs and tissues is a common treatment for organ failure. However, during the transplantation process, the donor organ or tissue is at risk for cell death since it is deprived of its normal blood

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supply prior to being implanted in the host. This ischemic state can be treated with cell death inhibitors by infusion into the donor organ or tissue, or by direct addition of the cell death inhibitors to the organ/tissue storage medium, i.e. bathing the organ or tissue in the storage medium containing the compound prior to transplant of the organ or tissue into a mammal. Cell death inhibitors may also be administered to animals to reduce or prevent cell death in the donor organ/tissue after it has been transplanted to protect it from the effects of host immune cells which kill their targets by triggering apoptosis. Examples of such organs and tissues include without limitation kidneys, livers, hearts, pancreases, lungs, corneas, hands, and embryonic nigral tissue.

The cytoprotective effects of cell death inhibitors can also be used to prevent the death of human or animal sperm and eggs used in *in vitro* fertilization procedures. These inhibitors can be used during the harvesting process and can also be included in the storage medium.

Mammalian cell lines, insect cells and yeast cells are commonly used to produce large amounts of recombinant proteins (such as antibodies, enzymes or hormones) for industrial or medicinal use. The lifespan of some of these cell lines is limited due to growth conditions, the nature of the recombinant molecule being expressed (some are toxic) and other unknown factors. The lifespans of industrial cell lines can be extended by including these cell death inhibitors in the growth medium in a concentration range of 1-100 μ M.

The factors governing hair growth and loss are largely unknown. There is some evidence, however, that hair follicle regression (referred to as catagen) may be due at least partially to apoptosis. Therefore, it is contemplated that the cell death inhibitors of the present invention can be used to treat hair loss that occurs due to various conditions, including but not limited to male-pattern baldness, radiation-induced or chemotherapy-induced hair loss, and hair loss due to emotional stress, by administering the inhibitors to the hair or hair follicles of an animal. There is also evidence that apoptosis may play a role in the loss of hair color. Therefore, it is contemplated that the

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cell death inhibitors of the present invention can also be used in treating or preventing cases of premature graying of the hair.

The death of skin epithelial cells can occur after exposure to high levels of radiation, heat or chemicals. It is contemplated that the cell death inhibitors of the present invention can be used to treat, reduce or prevent this type of skin damage. In one particular application, the cell death inhibitors can be applied as part of a topical formulation, e.g. an ointment, to the skin of an animal to treat acute over-exposure to the sun and to prevent blistering and peeling of the skin.

Goldberg *et al.* (*Nature Genetics* 13:442-449 (1996)) reported recently that huntingtin, a protein product of Huntington's disease (HD) gene, can be cleaved by CPP32 but not ICE. The mutation underlying HD is an expansion of a CAG trinucleotide at the 5' end of the HD gene. The trinucleotide expansion exceeding 36 repeats is associated with the clinical presentation of HD. The CAG expansion promotes cleavage of huntingtin by CPP32, thus links the role of CPP32 in the apoptotic cell death in HD. Compounds of the present invention with CPP32 inhibiting activity are useful in blocking CPP32 induced apoptotic cell death, thus in preventing and treating HD and other disorders characterized by expansion of trinucleotide repeats such as myotonic dystrophy, fragile X mental retardation, spinobulbar muscular atrophy, spinocerebellar atoxia type I and Dentato-Rubro pallidolusian atrophy.

The cell death inhibitors can also be used to treat or ameliorate small bowel tissue injury after intestinal ischemia-reperfusion, by administering to an animal in need thereof. See Farber, A. *et al.*, *J. Vascular Surg.* 30:752-760 (1999).

The invention relates to a method of treating, ameliorating or preventing oral mucositis, gastrointestinal mucositis, bladder mucositis, proctitis, bone marrow cell death, skin cell death and hair loss resulting from chemotherapy or radiation therapy of cancer in an animal, comprising administering to the animal in need thereof an effective amount of a caspase inhibitor.

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When animals are treated with chemotherapeutic agents and/or radiation to kill cancer cells, an unwanted side effect is the apoptotic death of rapidly dividing non-cancer cells. Such non-cancer cells include cells of the gastrointestinal tract, skin, hair, and bone marrow cells. According to the present invention, caspase inhibitors are administered to such non-cancer cells to prevent apoptosis of such cells. In a preferred embodiment, the caspase inhibitors are administered locally, e.g. to the gastrointestinal tract, mouth, skin or scalp to prevent apoptosis of the gastrointestinal, mouth, skin or hair cells but allowing for the death of the cancer cells. Thus, in one example, it is possible to treat brain cancer with chemotherapy or radiation therapy and protect the outer skin, hair cells, gastrointestinal tract and bone marrow by local administration of a caspase inhibitor. In the case of oral mucositis, the caspase inhibitor can be applied, for example, in the form of a mouth wash or mouth rinse, in a gel, or in the form of an oral slow release lozenge to prevent activation of caspases and apoptotic cell death induced by the chemotherapeutic agent or by radiation. In the case of gastrointestinal mucositis, the caspase inhibitor can be applied in a form such that it is not absorbed systemically or in a form that coats the surface of the gastrointestinal tract, or a suppository formulation for the treatment of gastrointestinal mucositis. In the case of proctitis, the caspase inhibitor may be applied as part of an enema or suppository. In the case of bladder mucositis, the caspase inhibitor may be applied through a bladder catheter. For prevention of radiation or chemotherapy-induced hair loss, the caspase inhibitor can be applied, for example, to the scalp in the form of a hair rinse, hair gel, shampoo or hair conditioner. Importantly, the caspase inhibitor can be applied prior to the administration of the chemotherapeutic agent or radiation, thus preventing or reducing the onset of the damaging effects of the chemotherapeutic agent or radiation to the normal cells.

Compositions within the scope of this invention include all compositions wherein the compounds of the present invention are contained in an amount which is effective to achieve its intended purpose. While

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individual needs vary, determination of optimal ranges of effective amounts of each component is within the skill of the art. Typically, the compounds may be administered to mammals, e.g. humans, orally at a dose of 0.0025 to 50 mg/kg, or an equivalent amount of the pharmaceutically acceptable salt thereof, per day of the body weight of the mammal being treated for apoptosis-mediated disorders, e.g., neuronal cell death, heart disease, retinal disorders, polycystic kidney disease, immune system disorders and sepsis. Preferably, about 0.01 to about 10 mg/kg is orally administered to treat or prevent such disorders. For intramuscular injection, the dose is generally about one-half of the oral dose. For example, for treatment or prevention of neuronal cell death, a suitable intramuscular dose would be about 0.0025 to about 25 mg/kg, and most preferably, from about 0.01 to about 5 mg/kg.

The unit oral dose may comprise from about 0.01 to about 50 mg, preferably about 0.1 to about 10 mg of the compound. The unit dose may be administered one or more times daily as one or more tablets each containing from about 0.1 to about 10, conveniently about 0.25 to 50 mg of the compound or its solvates.

In a topical formulation, the compound may be present at a concentration of about 0.01 to 100 mg per gram of carrier.

In addition to administering the compound as a raw chemical, the compounds of the invention may be administered as part of a pharmaceutical preparation containing suitable pharmaceutically acceptable carriers comprising excipients and auxiliaries which facilitate processing of the compounds into preparations which can be used pharmaceutically. Preferably, the preparations, particularly those preparations which can be administered orally or topically and which can be used for the preferred type of administration, such as tablets, dragees, slow release lozenges and capsules, mouth rinses and mouth washes, gels, liquid suspensions, hair rinses, hair gels, shampoos and also preparations which can be administered rectally, such as suppositories, as well as suitable solutions for administration by injection,

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topically or orally, contain from about 0.01 to 99 percent, preferably from about 0.25 to 75 percent of active compound(s), together with the excipient.

Also included within the scope of the present invention are the non-toxic pharmaceutically acceptable salts of the compounds of the present invention. Acid addition salts are formed by mixing a solution of the particular cell death inhibitors of the present invention with a solution of a pharmaceutically acceptable non-toxic acid such as hydrochloric acid, fumaric acid, maleic acid, succinic acid, acetic acid, citric acid, tartaric acid, carbonic acid, phosphoric acid, oxalic acid, and the like. Basic salts are formed by mixing a solution of the particular cell death inhibitors of the present invention with a solution of a pharmaceutically acceptable non-toxic base such as sodium hydroxide, potassium hydroxide, choline hydroxide, sodium carbonate Tris and the like.

The pharmaceutical compositions of the invention may be administered to any animal which may experience the beneficial effects of the compounds of the invention. Foremost among such animals are mammals, e.g., humans, although the invention is not intended to be so limited.

The caspase inhibitors and pharmaceutical compositions thereof may be administered by any means that achieve their intended purpose. For example, administration may be by parenteral, subcutaneous, intravenous, intramuscular, intraperitoneal, transdermal, buccal, intrathecal, intracranial, intranasal or topical routes. Alternatively, or concurrently, administration may be by the oral route. The dosage administered will be dependent upon the age, health, and weight of the recipient, kind of concurrent treatment, if any, frequency of treatment, and the nature of the effect desired. In general, the caspase inhibitors are administered locally to the tissues that are to be protected from apoptosis and separately from the chemotherapeutic agent. For example, cisplatin may be administered by i.v. injection to treat a cancer such as brain, lung, breast, liver, kidney, pancreatic, ovarian, prostatic cancer, and the caspase inhibitor administered locally to treat, ameliorate, or prevent apoptotic cell death in the mouth or gastrointestinal tract, such as a mouth wash

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for the treatment of oral mucositis; and IV injectable aqueous solution for the treatment of bone marrow cell death; and an oral formulation suitable for coating the gastrointestinal surfaces or an emema or suppository formulation for the treatment of gastrointestinal mucositis including proctitis. The caspase inhibitors may also be applied through a bladder catheter for the treatment, amelioration or prevention of bladder mucositis. Alternatively or concurrently, the caspase inhibitors may be applied topically to the skin and/or scalp to treat, ameliorate or prevent apoptotic cell death of hair and skin cells. In a further embodiment, the chemotherapeutic agent or radiation may be applied locally to treat a localized cancer such as brain, lung, breast, liver, kidney, pancreatic, ovarian, prostatic cancer, and the caspase inhibitor administered systemically, e.g. by i.v. injection, to treat, ameliorate or prevent apoptotic cell death of the gastrointestinal tract cells, mouth epithelial cells, bone marrow cells, skin cells and hair cells. In the case of oral mucositis in brain cancer treatment, for example, a caspase inhibitor that does not penetrate the blood-brain barrier can be applied, for example, systemically by i.v. injection followed by irradiation of the brain tumor. This would protect the oral mucosa from the harmful effects of radiation but the caspase inhibitor would not protect the brain tumor from the therapeutic effects of radiation. Importantly, the caspase inhibitor can be applied prior to administration of the radiation, thus preventing the onset of the damaging effects of the radiation to the normal mucosa cells.

The pharmaceutical preparations of the present invention are manufactured in a manner which is itself known, for example, by means of conventional mixing, granulating, dragee-making, dissolving, or lyophilizing processes. Thus, pharmaceutical preparations for oral use can be obtained by combining the active compounds with solid excipients, optionally grinding the resulting mixture and processing the mixture of granules, after adding suitable auxiliaries, if desired or necessary, to obtain tablets or dragee cores.

Suitable excipients are, in particular, fillers such as saccharides, for example lactose or sucrose, mannitol or sorbitol, cellulose preparations and/or

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calcium phosphates, for example tricalcium phosphate or calcium hydrogen phosphate, as well as binders such as starch paste, using, for example, maize starch, wheat starch, rice starch, potato starch, gelatin, tragacanth, methyl cellulose, hydroxypropylmethylcellulose, sodium carboxymethylcellulose, and/or polyvinyl pyrrolidone. If desired, disintegrating agents may be added such as the above-mentioned starches and also carboxymethyl-starch, cross-linked polyvinyl pyrrolidone, agar, or alginic acid or a salt thereof, such as sodium alginate. Auxiliaries are, above all, flow-regulating agents and lubricants, for example, silica, talc, stearic acid or salts thereof, such as magnesium stearate or calcium stearate, and/or polyethylene glycol. Dragee cores are provided with suitable coatings which, if desired, are resistant to gastric juices. For this purpose, concentrated saccharide solutions may be used, which may optionally contain gum arabic, talc, polyvinyl pyrrolidone, polyethylene glycol and/or titanium dioxide, lacquer solutions and suitable organic solvents or solvent mixtures. In order to produce coatings resistant to gastric juices, solutions of suitable cellulose preparations such as acetylcellulose phthalate or hydroxypropylmethyl-cellulose phthalate, are used. Dye stuffs or pigments may be added to the tablets or dragee coatings, for example, for identification or in order to characterize combinations of active compound doses.

Other pharmaceutical preparations which can be used orally include push-fit capsules made of gelatin, as well as soft, sealed capsules made of gelatin and a plasticizer such as glycerol or sorbitol. The push-fit capsules can contain the active compounds in the form of granules which may be mixed with fillers such as lactose, binders such as starches, and/or lubricants such as talc or magnesium stearate and, optionally, stabilizers. In soft capsules, the active compounds are preferably dissolved or suspended in suitable liquids, such as fatty oils, or liquid paraffin. In addition, stabilizers may be added.

Possible pharmaceutical preparations which can be used rectally include, for example, suppositories, which consist of a combination of one or more of the active compounds with a suppository base. Suitable suppository

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bases are, for example, natural or synthetic triglycerides, or paraffin hydrocarbons. In addition, it is also possible to use gelatin rectal capsules which consist of a combination of the active compounds with a base. Possible base materials include, for example, liquid triglycerides, polyethylene glycols, or paraffin hydrocarbons.

Suitable formulations for parenteral administration include aqueous solutions of the active compounds in water-soluble form, for example, water-soluble salts and alkaline solutions. In addition, suspensions of the active compounds as appropriate oily injection suspensions may be administered. Suitable lipophilic solvents or vehicles include fatty oils, for example, sesame oil, or synthetic fatty acid esters, for example, ethyl oleate or triglycerides or polyethylene glycol-400 (the compounds are soluble in PEG-400). Aqueous injection suspensions may contain substances which increase the viscosity of the suspension include, for example, sodium carboxymethyl cellulose, sorbitol, and/or dextran. Optionally, the suspension may also contain stabilizers.

In accordance with one aspect of the present invention, compounds of the invention are employed in topical and parenteral formulations and are used for the treatment of skin damage, such as that caused by exposure to high levels of radiation, including ultraviolet radiation, heat or chemicals.

One or more additional substances which have therapeutic effects on the skin may also be incorporated in the compositions. Thus, the composition may also contain one or more compounds capable of increasing cyclic-AMP levels in the skin. Suitable compounds include adenosine or a nucleic acid hydrolysate in an amount of about 0.1-1% and papaverine, in an amount of about 0.5-5%, both by weight based on the weight of the composition. Also suitable are β -adrenergic agonists such as isoproterenol, in an amount of about 0.1-2% or cyclic-AMP, in an amount of about 0.1-1%, again both by weight based on the weight of the composition. Other suitable types of additional active ingredients which may be incorporated in the compositions of this invention include any compounds known to have a beneficial effect on skin.

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Such compounds include retinoids such as Vitamin A, in an amount of about 0.003-0.3% by weight and chromanols such as Vitamin E or a derivative thereof in an amount of about 0.1-10% by weight, both based on the weight of the composition. Additionally, anti-inflammatory agents and keratoplastic agents may be incorporated in the cosmetic composition. A typical anti-inflammatory agent is a corticosteroid such as hydrocortisone or its acetate in an amount of about 0.25-5% by weight, or a corticosteroid such as dexamethasone in an amount of about 0.025-0.5% by weight, both based on the weight of the composition. A typical keratoplastic agent is coal tar in an amount of about 0.1-20% or anthralin in an amount of about 0.05-2% by weight, both based on the weight of the composition.

The topical compositions of this invention are formulated preferably as oils, creams, lotions, ointments and the like by choice of appropriate carriers. Suitable carriers include vegetable or mineral oils, white petrolatum (white soft paraffin), branched chain fats or oils, animal fats and high molecular weight alcohol (greater than C₁₂). The preferred carriers are those in which the active ingredient is soluble. Emulsifiers, stabilizers, humectants and antioxidants may also be included as well as agents imparting color or fragrance, if desired. Additionally, transdermal penetration enhancers can be employed in these topical formulations. Examples of such enhancers can be found in U.S. Patent Nos. 3,989,816 and 4,444,762.

Creams are preferably formulated from a mixture of mineral oil, self-emulsifying beeswax and water in which mixture the active ingredient, dissolved in a small amount of an oil such as almond oil, is admixed. A typical example of such a cream is one which includes about 40 parts water, about 20 parts beeswax, about 40 parts mineral oil and about 1 part almond oil.

Ointments may be formulated by mixing a solution of the active ingredient in a vegetable oil such as almond oil with warm soft paraffin and allowing the mixture to cool. A typical example of such an ointment is one

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which includes about 30% almond oil and about 70% white soft paraffin by weight.

Lotions may be conveniently prepared by dissolving the active ingredient, in a suitable high molecular weight alcohol such as propylene glycol or polyethylene glycol.

In addition, these compositions may include other medicinal agents, growth factors, wound sealants, carriers, etc., that are known or apparent to those skilled in the art. The compositions of the invention are administered to a warm-blooded animal, such as human, already suffering from a skin damage, such as a burn, in an amount sufficient to allow the healing process to proceed more quickly than if the host were not treated. Amounts effective for this use will depend on the severity of the skin damage and the general state of health of the patient being treated. Maintenance dosages over a prolonged period of time may be adjusted as necessary. For veterinary uses, higher levels may be administered as necessary.

In the case of an animal suffering from decreased hair growth, the compositions of the invention are administered in an amount sufficient to increase the rate of hair growth. Amounts effective for this use will depend on the extent of decreased hair growth, and the general state of health of the patient being treated. Maintenance dosages over a prolonged period of time may be adjusted as necessary. For veterinary uses, higher levels may be administered as necessary.

When the compounds are to be administered to plants, they may be applied to the leaves and/or stems and/or flowers of the plant, e.g. by spraying. The compounds may be sprayed in particulate form or dissolved or suspended in an appropriate carrier, e.g. in water or an oil-water emulsion. The compounds may also be combined with the soil of the plant. In this embodiment, the compounds are taken up by the roots of the plant.

In a preferred embodiment, the caspase inhibitor is formulated as part of a mouthwash for the treatment, amelioration or prevention of oral mucositis. Such mouthwashes are aqueous solutions of the caspase inhibitor

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which may also contain alcohol, glycerin, synthetic sweeteners and surface-active, flavoring and coloring agents. They may also contain anti-infective agents such as hexetidine and cetylpyridinium chloride. The mouthwashes may also contain topical anesthetics (e.g. benzocaine, cocaine, dyclonine hydrochloride, lidocaine, proparacaine hydrochloride or teracaine hydrochloride), for example, for relieving pain of radiation or chemotherapy-induced sores. The mouth washes may have either acidic or basic pH. See *Remington's Pharmaceutical Sciences*, A.R. Gennaro, ed., Mack Publishing Company (1990), pp. 1045, 1046, 1526 and 1965.

In another preferred embodiment, the caspase inhibitor is formulated as an oral formulation which is capable of coating the gastrointestinal surfaces for the treatment, amelioration or prevention of gastrointestinal mucositis. Examples of gastrointestinal mucositis include esophageal mucositis, gastric mucositis, and intestinal mucositis. Such formulations may comprise gastric antacids such as aluminum carbonate, aluminum hydroxide gel, bismuth subnitrate, bismuth subsalicylate, calcium carbonate, dihydroxyaluminum sodium carbonate, magaldrate, magnesium carbonate, magnesium hydroxide, magnesium oxide, sodium bicarbonate, milk of bismuth, dihydroxyaluminum aminoacetate, magnesium phosphate, magnesium trisilicate and mixtures thereof. Other additives include without limitation H₂-receptor antagonists, digestants, anti-emetics, adsorbants, and miscellaneous agents. See *Remington's Pharmaceutical Sciences*, A.R. Gennaro, ed., Mack Publishing Company (1990), pp. 774-778.

Chemotherapy agents such as cisplatin and radiation therapy often induce early and late onset emesis in the patient. Thus, in one embodiment an antiemetic is coadministered together with the caspase inhibitor to avoid emesis and retain contact of the caspase inhibitor with the gastrointestinal tract. Examples of such antiemetics include without limitation compounds that block the dopaminergic emetic receptors such as metoclopramide and trimethobenzamide, and cannabinoids. Metoclopramide may be administered orally prior to and/or during chemotherapy/radiation therapy/caspase inhibitor

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therapy to prevent the early emesis response and then later by intranasal administration according to U.S. Patent Nos. 5,760,086 and 4,536,386 to prevent delayed onset emesis. During the period after chemotherapy/radiation therapy, both the caspase inhibitor and the antiemetic may be coadministered to treat, ameliorate or prevent gastrointestinal mucositis.

In a further embodiment, the caspase inhibitor may be formulated as an IV injectable solution for the treatment, amelioration or prevention of bone marrow cell death.

The compositions of the compositions may be administered to a warm-blooded animal, such as human, already suffering from chemotherapy or radiation therapy-induced non-cancer cell death, or, more preferably, before or during therapy with chemotherapy or radiation.

The following examples are illustrative, but not limiting, of the method and compositions of the present invention. Other suitable modifications and adaptations of the variety of conditions and parameters normally encountered in clinical therapy and which are obvious to those skilled in the art are within the spirit and scope of the invention.

Example 1

S-1-(Carbonyl-Asp-CH₂F)ethyl N-Phenylcarbamate

Step A. *t*-Butyl 5-fluoro-3-[S-2-(phenylamino)carbonyloxypropionamido]-4-hydroxypentanoate. A mixture of *S*-(-)-2-[(phenylamino)carbonyloxy]propionic acid (201 mg, 0.96 mmol), EDCI (98 mg, 0.51 mmol), HOBT (74 mg, 0.48 mmol), DMAP (28 mg, 0.23 mmol) and *t*-butyl 3-amino-5-fluoro-4-hydroxypentanoate (101 mg, 0.49 mmol) in THF (6 mL) was stirred at room temperature for 17 h. It was diluted with 1:1 hexane/EtOAc (80 mL), washed with water, 2N HCl, water, saturated NaHCO₃, brine and dried over Na₂SO₄. The solution was evaporated and the residue was purified by flash chromatography (hexane/EtOAc 3/2) to give the title compound as a white solid (80 mg, 0.21 mmol, 43%). ¹H NMR (CDCl₃): 7.40-7.05 (m, 6H),

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5.18 (q, J = 6.9, 1H), 4.54-3.95 (m, 4H), 2.75-2.51 (m, 2H), 1.51 (d, J = 6.9, 3H), 1.42 (m, 9H).

Step B. *1-(Carbonyl-Asp(O-Bu-t)-CH₂F)ethyl N-phenylcarbamate.* A mixture of periodinane (0.49 g, 1.16 mmol) and t-butyl 5-fluoro-3-[S-2-(phenylamino)carbonyloxypropionamido]-4-hydroxypentanoate (80 mg, 0.21 mmol) in dichloromethane (6 mL) was refluxed for 20 h. After cooled to room temperature, a saturated sodium bicarbonate aqueous solution (10 mL) containing 0.5 g of Na₂S₂O₃ was added. The resulting mixture was stirred for 1 h, extracted with 1:1 hexane/EtOAc (80 mL). The organic phase was washed with water and brine, dried over sodium sulfate and concentrated *in vacuo* to yield the title compound as a hygroscopic white solid (69 mg, 0.18 mmol, 85%). ¹H NMR (CDCl₃): 7.40-7.25 (m, 5H), 7.13-7.00 (m, 2H), 5.30-4.87 (m, 4H), 3.05-2.71 (m, 2H), 1.56-1.53 (m, 3H), 1.42 (s, 9H).

Step C. *1-(Carbonyl-Asp-CH₂F)ethyl N-phenylcarbamate.* To a solution of 1-(carbonyl-Asp(O-Bu-t)-CH₂F)ethyl N-phenylcarbamate (69 mg, 0.18 mmol) in 3 mL of CH₂Cl₂ was added 1 mL of TFA. The resulting solution was stirred at room temperature for 1 h, diluted with EtOAc (80 mL), neutralized with saturated Na₂HPO₄ to pH~5, washed with water and brine, dried over Na₂SO₄ and concentrated *in vacuo* to give the title compound as a white solid (26 mg, 0.078 mmol, 44%). ¹H NMR (DMSO-*d*₆): 9.86 (br s, 1H), 8.67 (br s, 1H), 7.45-7.27 (m, 4H), 6.99 (br s, 1H), 5.30-4.80 (m, 4H), 4.59 (br s, 1H), 2.72 (br s, 2H), 1.37 (br s, 3H).

Example 2

2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-Phenylcarbamate

Step A. *2-Phenylaminocarbonyloxy-3-methylbutyric acid.* To a solution of 2-hydroxy-3-methylbutyric acid (0.20 g, 1.7 mmol) in pyridine (8 ml) was added phenyl isocyanate (0.184 mL). The resulting solution was stirred at room temperature for 4 h, acidified with 2N HCl to pH~2 and diluted with EtOAc (30 mL). The organic phase was isolated, washed with water and

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brine, dried over Na₂SO₄ and concentrated *in vacuo*. The residue was purified by chromatography (3:2 hexane/EtOAc then EtOAc) to yield the title compound as a colorless oil (200 mg, 0.85 mmol, 50%). ¹H NMR (DMSO-*d*₆): 9.80 (s, 1H), 7.47 (d, J = 7.8, 2H), 7.28 (t, J = 7.8, 2H), 7.00 (t, J = 7.8, 1H), 4.71 (d, J = 4.2, 1H), 2.17 (m, 1H), 1.02-0.97 (m, 6H).

The title compound was prepared in three steps as described in Example 1 from 2-phenylaminocarbonyloxy-3-methylbutyric acid and t-butyl-3-amino-5-fluoro-4-hydroxypentanoate. ¹H NMR (DMSO-*d*₆): 9.79 (s, 1H), 8.62 (s, 1H), 7.47 (d, J = 7.8, 2H), 7.28 (t, J = 7.8, 2H), 7.00 (t, J = 7.8, 1H), 5.14-4.60 (m, 4H), 2.80-2.60 (m, 2H), 2.10 (m, 1H), 0.97-0.93 (m, 6H).

Example 3

S-2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-Phenylcarbamate

The title compound was prepared in four steps as described in Examples 1 and 2 from S-(+)-2-hydroxy-3-methylbutyric acid and phenyl isocyanate. ¹H NMR (DMSO-*d*₆): 9.79 (s, 1H), 8.60 (s, 1H), 7.47 (d, J = 7.5, 2H), 7.28 (t, J = 8.2, 2H), 7.00 (t, J = 7.2, 1H), 5.14-4.58 (m, 4H), 2.80-2.71 (m, 2H), 2.10 (m, 1H), 0.98-0.93 (m, 6H).

Example 4

S-1-(Carbonyl-Asp-CH₂F)ethyl N-Benzylcarbamate

The title compound was prepared in four steps as described in Examples 1 and 2 from L-(+)-lactic acid and benzyl isocyanate. ¹H NMR (acetone-*d*₆): 7.81-7.00 (m, 7H), 5.37-4.33 (m, 6H), 2.89-2.67 (m, 2H), 1.40 (d, J = 6.9, 3H).

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Example 5***2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-Benzylcarbamate***

The title compound was prepared in four steps as described in Examples 1 and 2 from 2-hydroxy-3-methylbutyric acid and benzyl isocyanate. ¹H NMR (DMSO-*d*₆): 12.48 (s, 1H), 8.59-8.51 (m, 1H), 7.90-7.80 (m, 1H), 7.33-7.24 (m, 5H), 5.34-4.99 (m, 2H), 4.70-4.29 (m, 3H), 4.18 (d, J = 6.0, 2H), 2.85-2.53 (m, 2H), 2.01 (m, 1H), 0.92-0.87 (m, 6H).

Example 6***S-2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-Benzylcarbamate***

The title compound was prepared in four steps as described in Examples 1 and 2 from S-(+)-2-hydroxy-3-methylbutyric acid and benzyl isocyanate. ¹H NMR (DMSO-*d*₆): 8.46 (s, 1H), 7.86 (t, J = 6.3, 1H), 7.34-7.21 (m, 5H), 4.97-4.56 (m, 4H), 4.18 (d, J = 6.0, 2H), 2.68 (m, 2H), 2.02 (m, 1H), 0.91-0.86 (m, 6H).

Example 7***S,S-2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-(2-Methyl-1-methoxycarbonylpropyl)-carbamate***

Step A. *S,S-1-Carboxy-2-methylpropyl N-(2-methyl-1-methoxycarbonyl-propyl)-carbamate.* To a solution of S-(+)-2-hydroxy-3-methylbutyric acid (0.2 g, 1.69 mmol) in pyridine (10 ml) was added methyl-(S)-(-)-2-isocyanato-3-methylbutyrate (0.243 mL, 1.69 mmol). The resulting solution was then stirred at room temperature for 24 h, acidified with 2N HCl to pH~2 and diluted with EtOAc (30 mL). The organic phase was isolated, washed with water and brine, and concentrated *in vacuo*. The residue was dissolved in saturated NaHCO₃, washed with 4:1 hexane: EtOAc. The aqueous phase was acidified with 2N HCl to pH~2, extracted with ethyl

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acetate (2x30 mL). The organic layer was combined and washed with water and brine, dried over Na_2SO_4 and concentrated *in vacuo* to yield the title compound as a colorless oil (92 mg, 0.334 mmol, 20%). ^1H NMR ($\text{DMSO-}d_6$): 6.42 (d, $J = 8.7$, 1H), 4.08-3.99 (m, 2H), 3.62 (d, $J = 0.9$, 3H), 2.04-1.93 (m, 2H), 0.94-0.83 (m, 12H).

Step B. *S,S*-2-Methyl-1-(carbamyl-*N*-(*t*-butyl 5-fluoro-4-hydroxypentanoate-3-yl)propyl *N*-(2-methyl-1-methoxycarbonylpropyl)carbamate. A mixture of *S,S*-1-carboxy-2-methylpropyl *N*-(2-methyl-1-methoxycarbonylpropyl)carbamate (92 mg, 0.335 mmol), EDCI (66 mg, 0.335 mmol), HOBt (45 mg, 0.335 mmol), DMAP (20 mg, 0.17 mmol) and tert-butyl-3-amino-5-fluoro-4-hydroxypentanoate (69 mg, 0.335 mmol) in THF (10 mL) was stirred at room temperature for 20 h, filtered and concentrated *in vacuo*. The residue was purified by chromatography (3:2 hexane/ EtOAc) to yield the title compound as a colorless oil (20 mg, 0.043 mmol, 13%). ^1H NMR (CDCl_3): 6.98-6.86 (m, 1H), 5.50 (d, $J = 9.3$, 1H), 5.10-5.05 (m, 1H), 4.90 (d, $J = 4.2$, 1H), 4.50-4.00 (m, 5H), 3.76 (s, 3H), 2.74-2.48 (m, 2H), 2.33-2.13 (m, 2H), 1.44 (s, 9H), 0.99-0.93 (m, 12H).

Step C. *S,S*-2-Methyl-1-(carbonyl-Asp(OBu-*t*) CH_2F)propyl *N*-(2-methyl-1-methoxycarbonylpropyl)carbamate. A mixture of Dess-Martin reagent (0.2 g, 0.47 mmol) and *S,S*-2-methyl-1-(carbamyl-*N*-(*t*-butyl 5-fluoro-4-hydroxypentanoate-3-yl)propyl *N*-(2-methyl-1-methoxycarbonylpropyl)-carbamate (20 mg, 0.043 mmol) in dichloromethane (10 mL) was refluxed for 20 h, cooled to room temperature, and 25 mL of saturated sodium bicarbonate aqueous solution containing 0.2 g of $\text{Na}_2\text{S}_2\text{O}_3$ was added. The resulting mixture was stirred for 2 days, diluted with EtOAc (50 mL), washed with water and brine, dried over Na_2SO_4 and concentrated *in vacuo*. The residue was purified by chromatography (3:1 hexane/EtOAc) to yield the title compound as a colorless oil (12 mg, 0.026 mmol, 60%). ^1H NMR (CDCl_3): 7.25-7.10 (m, 1H), 5.52-5.44 (m, 1H), 5.31-4.84 (m, 4H), 4.34-4.25 (m, 1H), 3.77 (d, $J = 1.2$, 3H), 3.09-2.66 (m, 2H), 2.34-2.12 (m, 2H), 1.42 (d, $J = 1.8$, 9H), 1.00-0.88 (m, 12H).

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Step D. *S,S*-2-Methyl-1-(carbonyl-Asp-CH₂F)propyl *N*-(2-Methyl-1-methoxycarbonylpropyl)-carbamate. To a solution of *S,S*-2-methyl-1-(carbonyl-Asp(OBu-*t*)CH₂F)propyl *N*-(2-methyl-1-methoxycarbonylpropyl)-carbamate (12 mg, 0.026 mmol) in 1.5 mL of CH₂Cl₂ was added 0.5 mL of TFA. The resulting solution was allowed to stir at room temperature for 4 h, diluted with CHCl₃ (3 x 15 mL), concentrated *in vacuo* to give the title compound as a light yellow solid (10.25 mg, 0.025 mmol, 96%). ¹H NMR (acetone-*d*₆): 7.75-7.40 (m, 1H), 6.85-6.82 (m, 1H), 5.34-5.12 (m, 1H), 4.84-4.80 (m, 2H), 4.58-4.43 (m, 1H), 4.18-4.11 (m, 1H), 3.71 (s, 3H), 2.88-2.60 (m, 2H), 2.16-2.05 (m, 2H), 0.98-0.86 (m, 12H).

Example 8

S-2-Methyl-1-(carbonyl-Asp-CH₂DCB)propyl *N*-Phenylcarbamate

Step A. Benzyloxycarbonyl-Asp(OBu-*t*)-CH₂DCB. To a mixture of benzyloxycarbonyl-Asp(OBu-*t*)-CH₂Br (300 mg, 0.75 mm), 2,6-dichlorobenzoic acid (215 mg, 1.12 mm) and KF (217 mg, 3.75 mm) was added anhydrous DMF (5 ml). The mixture was stirred under Ar at room temperature for 12 hours, and then was diluted with ethyl acetate (30 ml). The solution was washed with saturated aqueous ammonium chloride solution, brine, and then dried over Na₂SO₄. The solvent was removed, and the residue was purified with flash column chromatography (Hexane:EtOAc, 2:1). The title compound was obtained as a colorless oil (296 mg, 0.67 mm, 89%). ¹H NMR (CDCl₃): 7.34 (m, 8H), 5.97(d, J = 8.7, 1H), 5.21 (d, J = 6.6, 2H), 5.16 (s, 2H), 4.70 (m, 1H), 2.90 (m, 2H), 1.28 (s, 9H).

Step B. *S*-2-Methyl-1-(carbonyl-Asp(OBu-*t*)-CH₂DCB)propyl *N*-phenylcarbamate. To a solution of benzyloxycarbonyl-Asp(OBu-*t*)-CH₂DCB (150 mg, 0.34 mm) in ethanol (20 ml) was added Pd/C (30 mg) and aqueous HCl (0.2 ml, 6N). The mixture was stirred under H₂ for 3 h, and then the Pd/C was removed by filtration. The ethanol solution was concentrated, and the residue was carried on for the next reaction. A mixture of *S*-2-phenylamino-

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carbonyloxy-3-methylbutyric acid (81 mg, 0.34 mmol), EDCI (65 mg, 0.34 mmol), HOBt (52 mg, 0.34 mmol), DMAP (24 mg, 0.20 mmol) and NH₂-Asp(OBu-t)-CH₂DCB-HCl in THF (10 mL) was stirred at room temperature for 16 h, and then diluted with ethyl acetate (30 ml). The solution was washed with aqueous NaOH (1 N), aqueous HCl (1 N), brine, and dried over Na₂SO₄. The solution was concentrated, and the residue was purified by flash column chromatography to give the title compound as a pale yellow solid (41 mg, 0.07 mm, 22%). ¹H NMR (CDCl₃): 7.28 (m, 8H), 5.05 (m, 3H), 4.67 (m, 1H), 2.88 (m, 2H), 2.31 (m, 1H), 1.42 (s, 9H), 0.91 (m, 6H).

Step C. *S-2-Methyl-1-(carbonyl-Asp-CH₂DCB)propyl N-phenylcarbamate.* To a solution of *2-Methyl-1-(carbonyl-Asp(OBu-t)-CH₂DCB)propyl N-phenylcarbamate* in CH₂Cl₂ (2 ml) was added TFA (1 ml). The mixture was stirred at 25°C under Ar for 12 h, and then was diluted with ethyl acetate (30 ml). The solution was washed with saturated aqueous Na₂HPO₄, and dried over Na₂SO₄. It was evaporated and the residue was dried under vacuum. The title compound was obtained as yellow color solid (18 mg, 0.03 mm, 47%). ¹H NMR (CDCl₃): 7.29 (m, 8H), 5.00 (m, 3H), 4.21 (m, 1H), 2.91 (m, 2H), 2.35 (m, 1H), 0.97 (m, 6H).

Example 9

S-2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-(3-Fluorophenyl)carbamate

The title compound was prepared in four steps as described in Example 1 and 2 from S-(+)-2-hydroxy-3-methylbutyric acid and 3-fluorophenyl isocyanate. ¹H NMR (DMSO-*d*₆): 10.09 (bs, 1H), 8.73 (m, 1H), 8.29 (m, 1H), 7.35 (m, 3H), 6.83 (s, 1H), 5.18 (m, 1H), 4.69 (m, 3H), 2.60 (m, 2H), 2.11 (m, 1H), 0.96 (m, 6H).

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Example 10***S-2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-(4-Fluorophenyl)carbamate***

5 The title compound was prepared in four steps as described in Example 1 and 2 from S-(+)-2-hydroxy-3-methylbutyric acid and 4-fluorophenyl isocyanate. ¹H NMR (DMSO-*d*₆): 9.73 (bs, 1H), 8.56 (bs, 1H), 7.33 (s, 2H), 7.00 (s, 2H), 5.05 (m, 1H), 4.55 (m, 3H), 2.52 (m, 2H), 1.97 (m, 1H), 0.81 (m, 6H).

10

Example 11***S-2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-(3,4-Difluorophenyl)carbamate***

15 The title compound was prepared in four steps as described in Example 1 and 2 from S-(+)-2-hydroxy-3-methylbutyric acid and 3,4-difluorophenyl isocyanate. ¹H NMR (DMSO-*d*₆): 10.09 (bs, 1H), 8.65 (bs, 1H), 7.56 (m, 1H), 7.37 (m, 1H), 7.24 (m, 1H), 5.05 (m, 1H), 4.65 (m, 3H), 2.73 (m, 2H), 2.11 (m, 1H), 0.95 (m, 6H).

Example 12

20

S-2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-(4-Phenoxyphenyl)carbamate

25 The title compound was prepared in four steps as described in Example 1 and 2 from S-(+)-2-hydroxy-3-methylbutyric acid and 4-phenoxyphenyl isocyanate. ¹H NMR (DMSO-*d*₆): 12.45 (bs, 1H), 9.84 (bs, 1H), 8.65 (m, 1H), 7.48 (d, J=8.7 Hz, 2H), 7.35 (dd, J=7.5, 8.4 Hz, 2H), 7.08 (dd, J=7.2, 7.5 Hz, 2H), 6.95 (m, 3H), 5.22 (m, 1H), 4.69 (m, 3H), 2.62 (m, 2H), 2.09 (m, 1H), 0.93 (m, 6H).

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Example 13***S-1-Cyclohexyl-1-(carbonyl-Asp-CH₂F)methyl N-Phenylcarbamate***

5 The title compound was prepared in four steps from (S)-(+)-hexahydromandelic acid and phenyl isocyanate by the method described in Example 1 and 2 as a white solid. ¹H NMR (DMSO-*d*₆): 9.76 (bs, 1H), 8.68 (bs, 1H), 7.44-7.27 (m, 4H), 6.99 (bs, 1H), 5.30-4.80 (m, 4H), 1.80-1.45 (m, 6H), 1.14 (bs, 5H).

10

Example 14***S-2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-(2,5-Dichlorophenyl)carbamate***

15 The title compound was prepared in four steps as described in Example 1 and 2 from S-(+)-2-hydroxy-3-methylbutyric acid and 2,5-dichlorophenyl isocyanate. ¹H NMR (DMSO-*d*₆): 9.39 (s, 1H), 8.69 (bs, 1H), 7.75 (d, J=6.6 Hz, 1H), 7.52 (d, J=8.4 Hz, 1H), 7.24 (d, J=8.7 Hz, 1H), 5.19 (m, 1H), 4.72 (m, 3H), 2.75 (m, 2H), 2.10 (m, 1H), 0.93 (m, 6H).

20

Example 15***S-2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-(2,4-Dichlorophenyl)carbamate***

25 The title compound was prepared in four steps as described in Example 1 and 2 from S-(+)-2-hydroxy-3-methylbutyric acid and 2,4-dichlorophenyl isocyanate. ¹H NMR (CDCl₃): 8.10 (dd, J=8.7, 3.9 Hz, 1H), 7.39 (m, 1H), 7.26 (m, 3H), 5.12 (m, 2H), 4.88 (m, 2H), 2.87 (m, 2H), 2.32 (m, 1H), 0.97 (m, 6H).

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Example 16***S-2-Methyl-1-(carbonyl-Asp-CH₂PTP)propyl N-Phenylcarbamate***

The title compound was prepared in 3 steps as described in Example 8 from benzyloxycarbonyl-Asp(OBu-t)-CH₂Br and 5-hydroxy-1-phenyl-3-(trifluoromethyl)pyrazole. ¹H NMR (CDCl₃): 7.72 (m, 1H), 7.32 (m, 10H), 7.17 (m, 2H), 5.08 (m, 3H), 4.86 (m, 1H), 2.81 (m, 2H), 2.31 (m, 1H), 1.03 (m, 6H).

Example 17***Enzyme Activity***

The activity of S-2-methyl-1-(carbonyl-Asp-CH₂F)propyl N-phenylcarbamate as an inhibitor of caspase-3 was measured in a fluorometric enzyme assay. Enzyme activity was measured using synthetic peptide substrates attached to a fluorogenic leaving group. Cleavage of the synthetic substrate by the enzyme results in a fluorescent signal which is read in a spectrofluorometer or in a fluorometric microtiter plate reader.

12 concentrations of the test compound ranging from 30 pM to 10 μM were tested in the enzyme assay. The enzyme reaction was conducted in the presence of 2 ng rCaspase 3 (purchased from PharMingen, a Becton division company, San Diego, CA), various concentrations of test compound, 10 μM caspase 3 substrate Ac-DEVD-AMC (SEQ ID NO:1, purchased from Quality Controlled Biochemicals, Inc. Hopkinton, MA) and caspase buffer (20 mM PIPES, 100 mM NaCl, 10 mM DTT, 1mM EDTA, 0.1 % CHAPS and 10 % sucrose, pH 7.2) in a total volume of 100 μL. The enzyme reaction was carried out in a 96-well plate and incubated at 37 °C for 30 minutes. The plate was then read with a fluorescence plate reader (EG&G WALLAG 1420-002) using excitation filter at 355nm / emission filter at 460 nm. The data were analyzed using GraphPrism software to give an IC₅₀ value of 17 nM for S-2-methyl-1-(carbonyl-Asp-CH₂F)propyl N-phenylcarbamate.

Example 18

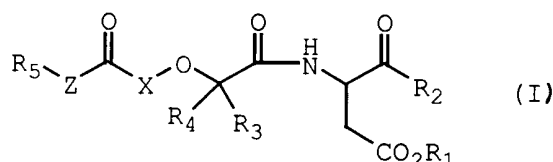
Cell Protection Activity

5 Cell protection activity was determined by examining the ability of *S*-2-methyl-1-(carbonyl-Asp-CH₂F)propyl *N*-phenylcarbamate to block HeLa cell death induced by TNF- α . In this assay, 25,000 HeLa cells are seeded in 96-well plates 24 hours prior to testing. On the assay day, the cells are pre-incubated for 30 minutes with the test compound at eight different
10 concentrations and the cells are then challenged with TNF- α (25 ng/ml) and cycloheximide (30 μ g/ml) and incubated further for 24 hours. The dead cells are then removed by two PBS washes and 100 μ L of a cell-mass indicator solution (calcein AM, final concentration 8 μ M) is added. After a 20 minute
15 incubation, the fluorescent signal is measured at 485/510 nm and the data are expressed as percent control, using control cultures treated only with cycloheximide. In these experiments, the EC₅₀ of *S*-2-methyl-1-(carbonyl-Asp-CH₂F)propyl *N*-phenylcarbamate for cytoprotection is 150 nM.

 Having now fully described this invention, it will be understood by those of ordinary skill in the art that the same can be performed within a wide
20 and equivalent range of conditions, formulations and other parameters without affecting the scope of the invention or any embodiment thereof. All patents, patent applications and publications cited herein are fully incorporated by reference herein in their entirety.

What Is Claimed Is:

1. A compound having the Formula I:



5

or a pharmaceutically acceptable salt or prodrug thereof, wherein

R₁ is an optionally substituted alkyl or hydrogen;

R₂ is hydrogen or optionally substituted alkyl;

10 R₃ and R₄ independently are hydrogen, optionally substituted aryl, optionally substituted heterocyclic, optionally substituted heteroaryl, optionally substituted carbocyclic, optionally substituted alkyl, optionally substituted alkenyl, or optionally substituted alkynyl;

15 R₅ is an optionally substituted alkyl, optionally substituted carbocyclic, optionally substituted heterocyclic, optionally substituted aryl or optionally substituted heteroaryl;

Z is O, S, NR₈, or (CR₉R₁₀)_n, where R₈, R₉ and R₁₀ independently are hydrogen, alkyl or cycloalkyl, and n is 0, 1, 2, or 3; and

X is a peptide of 1-2 amino acids or a bond.

20

2. The compound of claim 1, wherein R₃ and R₄ independently are hydrogen, aryl, heterocyclic, heteroaryl, C₁₋₁₀ alkyl, alkenyl, alkynyl, or C₁₋₁₀ alkyl substituted by one or more hydroxy, halogen, carboxy, amino, amide, ester, guanadino, thiol, alkylthiol, aryl, heterocyclic, or heteroaryl groups; and

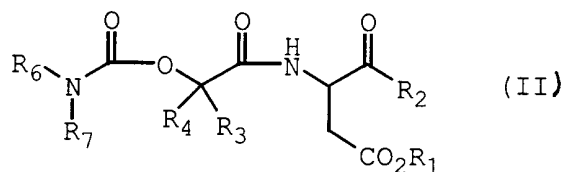
25 R₅ is an optionally substituted alkyl, C₄-C₇ cycloalkyl, saturated or unsaturated heterocyclic, aryl or heteroaryl group.

3. A compound according to claim 1, wherein R₁ is H, Me, Et or acetoxymethyl.

30

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4. A compound according to claim 1, wherein R_2 is hydrogen, fluoromethyl, acyloxymethyl, arylacyloxymethyl, aryloxymethyl, heteroaryloxymethyl, or aminomethyl.
5. A compound according to claim 1, wherein X is a bond.
6. A compound according to claim 1, wherein Z is O, S, NH or CH_2 .
7. A compound according to claim 1, wherein R_3 is hydrogen and R_4 is straight-chained or branched C_{1-6} alkyl, cycloalkyl, aryl or heteroaryl.
8. A compound according to claim 1, wherein R_3 is hydrogen and R_4 is straight-chained or branched C_{1-6} alkyl optionally substituted by hydroxy, halogen, carboxy, amino, amide, ester, guanadino, thiol, alkylthiol, aryl, heterocyclic or heteroaryl.
9. A compound according to claim 1, wherein R_5 is optionally substituted benzyl.
10. A compound according to claim 1, wherein R_5 is optionally substituted phenyl, naphthyl or heteroaryl.
11. A compound according to claim 1, wherein said compound has the Formula II:



or a pharmaceutically acceptable salt or prodrug thereof wherein

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R₆ and R₇ independently are hydrogen, alkyl, optionally substituted alkyl, C₄-C₇ cycloalkyl, heterocyclic, aryl, heteroaryl, or R₆ and R₇ are combined together with the nitrogen to form a heterocycle.

5 12. A compound according to claim 11, wherein R₂ is hydrogen, fluoromethyl, acyloxymethyl, arylacyloxymethyl, aryloxymethyl, heteroaryloxymethyl, or aminomethyl.

10 13. A compound according to claim 11, wherein R₁ is H, Me, Et or acetoxymethyl.

 14. A compound according to claim 11, wherein R₃ is hydrogen and R₄ is straight-chained or branched C₁₋₆ alkyl, cycloalkyl, aryl or heteroaryl.

15 15. A compound according to claim 11, wherein R₃ is hydrogen and R₄ is straight-chained or branched C₁₋₆ alkyl optionally substituted by hydroxy, halogen, carboxy, amino, amide, ester, guanadino, thiol, alkylthiol, aryl, heterocyclic or heteroaryl.

20 16. A compound according to claim 11, wherein R₆ is hydrogen and R₇ is optionally substituted phenyl, naphthyl, heteroaryl or benzyl.

 17. A compound according to claim 11, wherein R₆ is hydrogen and R₇ is an optionally substituted alkyl.

25 18. A compound according to claim 1, wherein said compound is selected from the group consisting of:

1-(Carbonyl-Asp-CH₂F)ethyl N-phenylcarbamate,

1-(Carbonyl-Asp-CH₂F)ethyl N-benzylcarbamate,

30 2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-phenylcarbamate,

2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-benzylcarbamate,

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2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-(2,6-dichlorophenyl)carbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-(2,5-dichlorophenyl)-carbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-(2,4-dichlorophenyl)-carbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂DCB)propyl N-phenylcarbamate,
 5 2-Methyl-1-(carbonyl-Asp-CH₂DCB)propyl N-(2,6-dichlorophenyl)-
 carbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂PTP)propyl N-phenylcarbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂PTP)propyl N-(2,6-dichlorophenyl)-carbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂DPP)propyl N-phenylcarbamate,
 10 2-Methyl-1-(carbonyl-Asp-CH₂DPP)propyl N-(2,6-dichlorophenyl)-
 carbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-(2-methyl-1-methoxycarbonyl-
 propyl)carbamate, and
 Z-Valine 2-methyl-1-(carbonyl-Asp-CH₂F)propyl ester.

15

19. A compound according to claim 1, wherein said compound is selected from the group consisting of:

2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-(3-fluorophenyl)carbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-(4-fluorophenyl)carbamate,
 20 2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-(3,4-difluorophenyl)carbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂F)propyl N-(4-phenoxyphenyl)carbamate,
 1-(Carbonyl-Asp-CH₂F)propyl N-phenylcarbamate,
 1-(Carbonyl-Asp-CH₂F)butyl N-phenylcarbamate,
 1-(Carbonyl-Asp-CH₂F)-2-propenyl N-phenylcarbamate,
 25 2-(4-Imidazolyl)-1-(carbonyl-Asp-CH₂F)ethyl N-phenylcarbamate,
 2-Phenyl-1-(carbonyl-Asp-CH₂F)ethyl N-phenylcarbamate,
 2-Methyl-1-(carbonyl-Asp-CH₂F)butyl N-phenylcarbamate,
 3-Methyl-1-(carbonyl-Asp-CH₂F)butyl N-phenylcarbamate,
 1-Phenyl-1-(carbonyl-Asp-CH₂F)methyl N-phenylcarbamate,
 30 1-(2-Chlorophenyl)-1-(carbonyl-Asp-CH₂F)methyl N-phenylcarbamate,
 1-(4-Chlorophenyl)-1-(carbonyl-Asp-CH₂F)methyl N-phenylcarbamate,

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1-Cyclohexyl-1-(carbonyl-Asp-CH₂F)methyl N-phenylcarbamate,
2-Chloro-1-(carbonyl-Asp-CH₂F)ethyl N-phenylcarbamate, and
2,2,2-trifluoro-1-(carbonyl-Asp-CH₂F)ethyl N-phenylcarbamate.

5 20. A pharmaceutical composition, comprising a compound of
claim 1, and a pharmaceutically acceptable carrier.

 21. A method of inhibiting cell death of a cell or tissue, comprising
contacting said cell or tissue with an effective amount of a compound of
10 claim 1.

 22. A method of treating or ameliorating cell death in the central or
peripheral nervous system, retinal neurons, cardiac muscle or immune system
cells of an animal, comprising administering to the animal in need of such
15 treatment or ameliorating an effective amount of a compound of claim 1.

 23. The method of claim 22, wherein said cell death is in the
central or peripheral nervous system, and is due to one of:

 (a) a condition of ischemia and excitotoxicity selected from the group
20 consisting of focal ischemia due to stroke and global ischemia due to cardiac
arrest;

 (b) traumatic injury;

 (c) viral infection;

 (d) radiation-induced nerve cell death;

25 (e) a neurodegenerative disorder selected from the group consisting of
Alzheimer's disease, Parkinson's Disease, a prion disease, multiple sclerosis,
amyotrophic lateral sclerosis, and spinobulbar atrophy;

 (f) spinal cord injury; or

 (g) acute bacterial meningitis.

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24. The method of claim 22, wherein said cell death is in the central or peripheral nervous system, and is due to expansion of trinucleotide repeats of the Huntington's Disease gene.

5 25. The method of claim 22, wherein said cell death is due to Huntington's Disease.

26. The method of claim 22, wherein said cell death is in cardiac muscle tissue, and is due to myocardial infarction, congestive heart failure,
10 cardiomyopathy or viral infection of the heart.

27. The method of claim 22, wherein said cell death is in retinal neurons and is due to increased intraocular pressure, age-related macular degeneration or retinitis pigmentosa.

15 28. The method of claim 22, wherein said cell death is in the immune system, and is due to an immune deficiency disorder selected from the group consisting of acquired immune deficiency syndrome, severe combined immune deficiency syndrome and radiation-induced immune
20 suppression.

29. The method of claim 22, wherein said cell death is due to an autoimmune disorder selected from the group consisting of lupus erythematosus, rheumatoid arthritis and type I diabetes.

25 30. The method of claim 22, wherein said cell death is due to type I diabetes.

31. A method of treating or preventing polycystic kidney disease, renal amyloidosis, acute renal failure, cyclosporine A induced murin tubular
30 epithelial cell death, HIV-induced nephropathy or anemia/erythropoiesis in an

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animal, comprising administering to the animal in need of such treatment or preventing an effective amount of a compound of claim 1.

5 32. A method of protecting a mammalian organ or tissue from cell death due to deprivation of normal blood supply, comprising contacting said organ or tissue with an effective amount of a compound of claim 1.

 33. The method of claim 32, wherein said organ or tissue is present in a storage medium prior to transplant into a mammal.

10

 34. The method of claim 32, wherein said tissue is embryonic nigral tissue.

 35. The method of claim 32, wherein said contacting comprises infusion of said compound into the organ or tissue, or bathing of said organ or tissue in a storage medium which comprises said compound.

15

 36. A method of reducing or preventing cell death in a donor organ or tissue after it has been transplanted into a host due to the effects of host immune cells, comprising administering to said host in need thereof an effective amount of a compound of claim 1.

20

 37. A method of reducing or preventing the death of mammalian sperm or eggs used in *in vitro* fertilization procedures, comprising contacting said sperm or egg with an effective amount of a compound of claim 1.

25

 38. A method of extending the lifespan of a mammalian or yeast cell line, comprising contacting said cell line with a compound of claim 1.

30

 39. The method of claim 38, wherein said contacting comprises including said compound in a cell growth medium.

40. A method of treating or ameliorating hair loss or premature graying of the hair in a mammal, comprising contacting the hair or hair follicles of the mammal in need thereof with a compound of claim 1.

5

41. The method of claim 40, wherein hair loss is treated, and said hair loss is due to male-pattern baldness, radiation, chemotherapy or emotional stress.

10

42. A method of treating or ameliorating skin damage of a mammal due to exposure to high levels of radiation, heat or chemicals, comprising applying to the skin of the mammal in need thereof with a compound of claim 1.

15

43. The method of claim 42, wherein said compound is applied as part of an ointment.

20

44. The method of claim 42, wherein said skin damage is due to acute over-exposure to the sun, and wherein said treating reduces blistering and peeling of the skin.

25

45. A method of treating or ameliorating sepsis in an animal, comprising administering to the animal in need thereof an effective amount of a compound of claim 1.

46. A method of treating or ameliorating hepatitis in an animal, comprising administering to the animal in need thereof an effective amount of a compound of claim 1.

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47. A method of treating or ameliorating hereditary tyrosinemia type 1 in an animal, comprising administering to the animal in need thereof an effective amount of a compound of claim 1.

5 48. A method of treating or ameliorating chronic alcohol ingestion induced buccal mucosa cell death in an animal, comprising administering to the animal in need thereof an effective amount of a compound of claim 1.

10 49. A method of treating or ameliorating cell death in plants or flowers, comprising administering to the plants or flowers in need thereof an effective amount of a compound of claim 1.

15 50. A method of treating or ameliorating radiation or ultraviolet-irradiation induced cell death in an animal, comprising administering to the animal in need thereof an effective amount of a compound of claim 1.

20 51. A method of treating or ameliorating apoptotic death of bone marrow cells in myelodysplastic syndromes (MDS), comprising administering to the animal in need thereof an effective amount of a compound of claim 1.

52. A method of treating or ameliorating apoptotic cell death in acute pancreatitis, comprising administering to the animal in need thereof an effective amount of a compound of claim 1.

25 53. A method of treating or preventing the inflammatory response in psoriasis or inflammatory bowel disease, comprising administering to the animal in need thereof an effective amount of a compound of claim 1.

30 54. A method of treating or ameliorating organ apoptosis after burn injury, comprising administering to the animal in need thereof an effective amount of a compound of claim 1.

55. A method of treating or ameliorating small bowel tissue injury after intestinal ischemia-reperfusion, comprising administering to the animal in need thereof an effective amount of a compound of claim 1.

5

56. A method of treating, ameliorating or preventing oral mucositis, gastrointestinal mucositis, bladder mucositis, proctitis, bone marrow cell death, skin cell death or hair loss resulting from chemotherapy or radiation therapy of cancer in an animal, comprising administering to the animal in need thereof an effective amount of a compound of claim 1.

10

57. The method of claim 56, wherein said compound is administered topically or orally.

15

58. The method of claim 57, wherein said compound is formulated as part of a mouthwash for the treatment, amelioration or prevention of oral mucositis.

20

59. The method of claim 57, wherein said compound is formulated as part of a slow release buccal lozenge.

60. The method of claim 57, wherein said compound is formulated as part of a suppository.

25

61. The method of claim 57, wherein said compound is formulated as part of a gel.

30

62. The method of claim 56, wherein said compound is administered through a bladder catheter for the treatment, amelioration or prevention of bladder mucositis.

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63. The method of claim 56, wherein said compound is administered as part of an enema for the treatment, amelioration or prevention of proctitis.

5 64. The method of claim 56, wherein said compound is formulated as an oral formulation which is capable of coating the gastrointestinal surfaces for the treatment, amelioration or prevention of gastrointestinal mucositis.

10 65. The method of claim 64, wherein said gastrointestinal mucositis is esophageal mucositis, gastric mucositis, or intestinal mucositis.

15 66. The method of claim 56, wherein said compound is administered by i.v. injection for the treatment, amelioration or prevention of bone marrow cell death.

 67. The method of claim 56, wherein said compound is administered as part of a pharmaceutical composition comprising a pharmaceutically acceptable carrier.

20 68. The method of claim 56, wherein said compound is administered after chemotherapy or radiation therapy of cancer in said animal.

25 69. The method of claim 56, wherein said compound is administered during chemotherapy or radiation therapy of cancer in said animal.

 70. The method of claim 56, wherein said compound is administered prior to chemotherapy or radiation therapy of cancer in said animal.

-1-

SEQUENCE LISTING

<110> Cytovia Inc.

Wang, Yan

Cai, Sui

Weber, Eckard

Mills, Gordon

Green, Douglas

Guan, Lufeng

<120> Substituted alpha-Hydroxy Acid Caspase Inhibitors and the Use Thereof

<130> 1735.040PC02

<140> To Be Assigned

<141> To Be Assigned

<150> 60/157,077

<151> 1999-08-27

<150> 60/158,373

<151> 1999-10-12

<160> 1

<170> PatentIn version 3.0

<210> 1

<211> 7

<212> PRT

<213> UNSURE

<400> 1

Asp Glu Val Asp

1

INTERNATIONAL SEARCH REPORT

International Application No

PCT/US 00/23566

A. CLASSIFICATION OF SUBJECT MATTER

IPC 7 C07C271/28 C07C271/12 C07C271/22 A61K31/195

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

IPC 7 C07C A61K

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practical, search terms used)

EPO-Internal, CHEM ABS Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category *	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	WO 96 03982 A (SANOFI WINTHROP INC) 15 February 1996 (1996-02-15) cited in the application abstract; claims 1-9 ---	1-70
A	US 5 585 357 A (CHATURVEDULA PRASAD V ET AL) 17 December 1996 (1996-12-17) cited in the application claims 1-24 ---	1-70
A	EP 0 618 223 A (SANDOZ LTD ;SANDOZ AG (DE); SANDOZ AG (AT)) 5 October 1994 (1994-10-05) cited in the application page 10 -page 18; claims 1-10 ---	1-70
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☒ Further documents are listed in the continuation of box C.

☒ Patent family members are listed in annex.

* Special categories of cited documents:

A document defining the general state of the art which is not considered to be of particular relevance

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O document referring to an oral disclosure, use, exhibition or other means

P document published prior to the international filing date but later than the priority date claimed

T later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

X document of particular relevance: the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

Y document of particular relevance: the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art.

G document member of the same patent family

Date of the actual completion of the international search

5 December 2000

Date of mailing of the international search report

12/12/2000

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INTERNATIONAL SEARCH REPORT

International Application No
PCT/US 00/23566

C.(Continuation) DOCUMENTS CONSIDERED TO BE RELEVANT		
Category *	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
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